



École nationale des ponts et chaussées

Internship Report CERMICS

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Internship Summary Sheet

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I would also like to thank the entire CERMICS team for their warm welcome and availability, which made this experience even more enjoyable. Through this internship, I had the opportunity to speak with second-year master's interns, PhD students and postdoctoral researchers. This was highly rewarding, both mathematically and personally. I was also able to discover what life is like in a mathematics research laboratory.

Abstract

My internship took place at CERMICS, the applied mathematics laboratory of the École nationale des ponts et chaussées. The subject of my internship concerned the state spaces of multifactor Markov processes associated with Volterra processes. Aurélien Alfonsi gave me a paper to read on this topic, namely [7]. The initial objective was to understand this paper. I therefore spent time studying its proofs and carrying out simulations.

The paper mainly established an invariant-domain result for multifactor Markov processes associated with Volterra processes for a certain type of kernel. We first sought to extend this result to another type of kernel through a limiting argument. More generally, Aurélien Alfonsi asked me to investigate whether some results in the paper could be extended to slightly more general settings. This occupied most of my internship. I had the opportunity to read several papers and lecture notes in an attempt to answer these questions.

Introduction

This report presents the introductory research work that I carried out during my internship at CERMICS under the supervision of Aurélien Alfonsi. It is a mathematical report. Most of the work I completed during this internship is included here.

I chose to undertake an internship in a mathematics laboratory for several reasons. First, next year I will continue into the second-year master's programme in applied mathematics, Probabilities and Finance. From this perspective, a stochastic calculus internship seemed particularly relevant. I would like to work in quantitative finance after completing my studies, and it seems valuable to acquire as much mathematical knowledge as possible in order to work in this field. I am also open to the idea of pursuing a PhD after the second-year master's programme. It was therefore useful for me to gain an insight into mathematical research and determine whether doctoral study might appeal to me. Finally, as I did not take a gap year, I needed to complete a short summer internship, for which a mathematics internship was very well suited.

In the conclusion, I will reflect on this experience and explain what the internship contributed to my studies.

Presentation of CERMICS

CERMICS (Centre d’Enseignement et de Recherche en Mathématiques et Calcul Scientifique) is the applied mathematics laboratory of École des Ponts ParisTech. It stands out for its ability to connect deterministic and stochastic approaches, producing both theoretical advances and practical numerical tools.

The laboratory works closely with numerous industrial partners from a wide range of sectors (transport, energy, finance, engineering and the pharmaceutical industry), while maintaining strong links with academia in related disciplines such as biology, chemistry, physics and materials science.

Research at CERMICS is organised around three main areas:

- **Applied Probability:** focused on risk modelling and numerical stochastic methods.
- **Modelling, Analysis and Simulation:** this area develops tools for mathematical analysis, modelling and simulation, with applications including materials science, quantum chemistry and mechanics.
- **Optimisation:** dedicated to discrete and stochastic optimisation, as well as operations research.

In practice, CERMICS is spread over two floors: the **Applied Probability** and **Optimisation** groups are located on the second floor, while the **Modelling, Analysis and Simulation** group occupies the third floor. I therefore completed my internship within the **Applied Probability** team.

The atmosphere at CERMICS is particularly friendly. PhD students, interns and postdoctoral researchers enjoy very good working relationships.

Subject of My Internship

This document summarises my understanding of the paper [7], which served as the starting point for my internship work. I explain its content in my own words and provide details on some of its points. I highlight my own reasoning and investigate how certain results might be improved or extended. To this end, I performed many calculations in simple cases.

Whenever a result was not proved by me, this is stated and a reference is provided. The proofs I produced are often extensions based on the same approach as the original proof; this is also indicated whenever applicable. All graphs in this work come from my own numerical simulations. I endeavoured to write this report without reproducing existing material, since doing so would add little value. I have tried to emphasise my personal contributions and provide my own explanations.

However, I naturally had to discuss results that already exist. In such cases, I tried to highlight the difficult parts of the proof and add details that were absent from the paper containing it. Some sections (such as 2.3, 2.4, 2.5, 3.6.6) present my “own” results. The simulations in Section 3 are mine, and some applications are my own. Of course, all the topics I wrote about and explored during this internship were suggested to me by Aurélien Alfonsi.

The paper [7] starts from a solution of the Volterra equation and studies the associated multifactor Markov process. Its main result is the identification of an invariant domain

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for this multifactor Markov process. The class of domains obtained is a class of cones. Identifying the invariant domains of this multifactor Markov process has several applications.

1 Introduction

We work on a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t), \mathbb{P})$ carrying an $(\mathcal{F}_t)$ -Brownian motion $(W_t)_{t \geq 0}$. The starting point of the paper [7] is the stochastic Volterra equation

$$Y_t = Y_0 + \int_0^t K(t-s)b(Y_s)ds + \int_0^t K(t-s)\sigma(Y_s)dW_s \quad (1)$$

where the kernel K is called a Prony series and is given by

$$K(t) = \sum_{i=1}^N \gamma_i e^{-\rho_i t} \quad \text{with } 0 < \rho_1 \dots < \rho_N \text{ and } \forall i \in \llbracket 1, N \rrbracket, \gamma_i > 0.$$

A solution Y of the Volterra equation can be written as $Y_t = \sum_{i=1}^N \gamma_i Y_t^{(i)}$, where $Y_t^{(i)}$ satisfies the stochastic differential equation

$$dY_t^{(i)} = -\rho_i(Y_t^{(i)} - y_0^{(i)})dt + b(Y_t)dt + \sigma(Y_t)dW_t, \quad Y_0^{(i)} = y_0^{(i)} \quad (2)$$

with $\sum_{i=1}^N \gamma_i y_0^{(i)} = Y_0$ ¹. Indeed, integration by parts yields

$$\begin{aligned} d\left(e^{\rho_i t}(Y_t^{(i)} - y_0^{(i)})\right) &= \rho_i e^{\rho_i t} \left(Y_t^{(i)} - y_0^{(i)}\right) dt + e^{\rho_i t} dY_t^{(i)} \\ &= \rho_i e^{\rho_i t} \left(Y_t^{(i)} - y_0^{(i)}\right) dt \\ &\quad + e^{\rho_i t} \left(-\rho_i(Y_t^{(i)} - y_0^{(i)})dt + b(Y_t)dt + \sigma(Y_t)dW_t\right) \\ &= e^{\rho_i t} (b(Y_t)dt + \sigma(Y_t)dW_t). \end{aligned}$$

Thus, we obtain

$$Y_t^{(i)} = y_0^{(i)} + e^{-\rho_i t}(Y_0^{(i)} - y_0^{(i)}) + \int_0^t e^{-\rho_i(t-s)}(b(Y_s)ds + \sigma(Y_s)dW_s).$$

Therefore, summing gives

$$\sum_{i=1}^N \gamma_i Y_t^{(i)} = \sum_{i=1}^N \gamma_i y_0^{(i)} + \sum_{i=1}^N \gamma_i e^{-\rho_i t}(Y_0^{(i)} - y_0^{(i)}) + \int_0^t K(t-s)(b(Y_s)ds + \sigma(Y_s)dW_s).$$

Then, recalling that $\sum_{i=1}^N \gamma_i y_0^{(i)} = Y_0$ and $Y_0^{(i)} = y_0^{(i)}$, we have

$$\sum_{i=1}^N \gamma_i Y_t^{(i)} = Y_0 + \int_0^t K(t-s)(b(Y_s)ds + \sigma(Y_s)dW_s) = Y_t.$$

We write $\mathbf{Y}_t = \left(Y_t^{(i)}\right)_{i \in \llbracket 1, N \rrbracket}$, so that $Y_t = \gamma^\top \mathbf{Y}_t$. We also write $\mathbf{y}_0 = \left(y_0^{(i)}\right)_{i \in \llbracket 1, N \rrbracket}$. More generally, all vectors are written in bold. The N equations (2) can then be rewritten as the N -dimensional equation

$$d\mathbf{Y}_t = -\text{diag}(\boldsymbol{\rho})(\mathbf{Y}_t - \mathbf{y}_0)dt + b(\gamma^\top \mathbf{Y}_t)\mathbf{1}dt + \sigma(\gamma^\top \mathbf{Y}_t)\mathbf{1}dW_t. \quad (3)$$

In [7], the aim is to find domains $\mathcal{D} \subset \mathbb{R}^N$ such that, if $\mathbf{Y}_0 \in \mathcal{D}$, then there exists a weak solution $\mathbf{Y}$ of (3), starting from $\mathbf{Y}_0$, that remains in $\mathcal{D}$, in the sense that $\mathbf{Y}_t \in \mathcal{D}$ for every

¹In [7], it is stated that one may take $Y_0^i \neq y_0^i$, but if $\sum_{i=1}^N \gamma_i Y^{(i)}$ is to be the solution of the Volterra equation, the conditions I have given are indeed necessary, as shown by the calculation following this footnote.

$t \geq 0$ almost surely.

Assuming that, when $Y_0 \geq 0$, every solution Y of (1) starting from Y_0 satisfies $Y_t \geq 0$ for every $t \geq 0$, it follows that such a domain $\mathcal{D}$ must be contained in the half-space $\{\mathbf{y} \in \mathbb{R}^N \mid \bar{\gamma}^\top \mathbf{y} \geq 0\}$.

2 Main Result

The paper [7] provides a suitable class of domains $\mathcal{D}$. To describe them, we introduce a set of matrices.

2.1 Admissible Matrices

Definition 2.1 (Set of admissible matrices). *A matrix $Q \in \mathbb{R}^{N \times N}$ is said to be admissible if*

1. Q is invertible;
2. $\mathbf{e}_N^\top Q = \bar{\gamma}^\top$
3. $Q\mathbf{1} = \bar{\gamma}\mathbf{e}_N$, where $\bar{\gamma} := \sum_{i=1}^N \gamma_i$;
4. $(Q\text{diag}(\boldsymbol{\rho})Q^{-1})_{i,j} \leq 0 \quad \forall i \neq j \in \llbracket 1, N \rrbracket$

We denote the set of admissible matrices by $\mathcal{Q}$.

Remark 2.1. *Property 2 means that the last row of Q is $\bar{\gamma}^\top$, and Property 3 means that the sum of the entries in each row of Q is zero, except in the last row, where this sum is $\bar{\gamma}$.*

The paper [7] shows that $\mathcal{Q} \neq \emptyset$. Indeed, in dimension N , the matrix $Q = (q_{ij})$ defined by

$$q_{ij} = \begin{cases} \gamma_j & \text{if } j \leq i, \\ -\sum_{j=1}^i \gamma_j & \text{if } j = i+1 \text{ and } i \in \llbracket 1, N-1 \rrbracket, \\ 0 & \text{otherwise.} \end{cases} \quad (4)$$

is admissible.

One may ask whether infinitely many such matrices exist. The paper [7] answers this question when $N = 2$. In this case, the admissible matrices are those of the form

$$Q = \begin{pmatrix} q & -q \\ \gamma_1 & \gamma_2 \end{pmatrix}$$

with $q > 0$. Indeed, Properties 2 and 3 are immediate. Property 1 holds because $\det(Q) = \bar{\gamma}q$. Finally, a calculation shows that

$$Q\text{diag}(\boldsymbol{\rho})Q^{-1} = \frac{1}{\bar{\gamma}} \begin{pmatrix} \gamma_1\rho_2 + \gamma_2\rho_2 & (\rho_1 - \rho_2)q \\ \gamma_1\gamma_2(\rho_1 - \rho_2)q^{-1} & \gamma_1x_1 + \gamma_2x_2 \end{pmatrix}, \quad (5)$$

which shows that Property 4 holds.

When $N = 3$, the paper [7] looks for admissible matrices among those of the form

$$Q = \begin{pmatrix} 1 & -a & -1+a \\ 1 & b & -1-b \\ \gamma_1 & \gamma_2 & \gamma_3 \end{pmatrix}. \quad (6)$$

The paper [7] shows that Q is admissible if $a = 1$ and $b = \frac{\gamma_2}{\gamma_1} \neq 1$. We now show that there are in fact infinitely many admissible matrices of the preceding form.

For a and b satisfying $a + b \neq 0$, the matrix Q is invertible because

$$\det(Q) = (a + b)(\gamma_1 + \gamma_2 + \gamma_3).$$

Under this assumption, writing $R = -\bar{\gamma}Q\text{diag}(\boldsymbol{\rho})Q^{-1}$, $y_1 := \rho_2 - \rho_1$ and $y_2 = \rho_3 - \rho_2$, we can compute the coefficients r_{ij} of R ; it remains to impose the condition

$$r_{ij} \leq 0 \quad \forall i \neq j \in \llbracket 1, 3 \rrbracket.$$

According to [7], $r_{23} \leq 0$ always holds, and the other conditions $r_{ij} \leq 0$ can be written as

$$\begin{aligned} \text{(i)} \quad & a \leq \frac{y_1 + y_2}{y_2} \\ \text{(ii)} \quad & a \geq \frac{\gamma_2}{\gamma_1} \cdot \frac{\gamma_1 y_1 - \gamma_3 y_2}{\gamma_2 y_1 + \gamma_3(y_1 + y_2)} \\ \text{(iii)} \quad & b \geq \frac{\gamma_2}{\gamma_1} \cdot \frac{-\gamma_1 y_1 + \gamma_3 y_2}{\gamma_2 y_1 + \gamma_3(y_1 + y_2)} \\ \text{(iv)} \quad & 0 \leq \gamma_1 y_2 a^2 + ca - \gamma_2(y_1 + y_2) \\ \text{(v)} \quad & 0 \geq \gamma_1 y_2 b^2 - cb - \gamma_2(y_1 + y_2) \end{aligned}$$

where $c = \gamma_3 y_1 + \gamma_2(y_1 + y_2) - \gamma_1 y_2$.

If we take $\boldsymbol{\rho} = (\rho_1, \rho_1 + a^2, \rho_1 + a^2 + \varepsilon)$ with $1 > \varepsilon > 0$ and $a \geq 1$, then $y_1 = a^2$ and $y_2 = \varepsilon$. Hence

$$\frac{y_1 + y_2}{y_2} = \frac{a^2 + \varepsilon}{\varepsilon} \geq a^2$$

and therefore Condition (i) holds if $a \geq 1$. Next, the function φ defined by

$$\varphi(a) := \frac{\gamma_1 y_1 - \gamma_3 y_2}{\gamma_2 y_1 + \gamma_3(y_1 + y_2)} = \frac{\gamma_1 a^2 - \gamma_3 \varepsilon}{\gamma_2 a^2 + \gamma_3(a^2 + \varepsilon)}$$

is bounded, so the quantities

$$K = \inf_{x \geq 0} \varphi(x) \quad \text{and} \quad K' = \sup_{x \geq 0} \varphi(x).$$

are finite.

Moreover,

$$\varphi'(a) = \frac{2\varepsilon a \Delta}{((\gamma_2 + \gamma_3)a^2 + \gamma_3 \varepsilon)^2} \tag{7}$$

where $\Delta = \gamma_1 \gamma_2 + \gamma_3(\gamma_2 + \gamma_3) > 0$. Thus φ is strictly increasing on $\mathbb{R}_+$ and

$$K = \varphi(0) = -1 \quad \text{and} \quad K' = \lim_{a \rightarrow \infty} \varphi(a) = \frac{\gamma_1}{\gamma_2 + \gamma_3}$$

Thus, for $a \geq \frac{\gamma_2}{\gamma_1} K'$, Condition (ii) holds, and for $b \geq \frac{\gamma_2}{\gamma_1} K$, Condition (iii) holds. Moreover,

$$\gamma_1 y_2 a^2 + ca - \gamma_2(y_1 + y_2) = (\gamma_2 + \gamma_3)a^3 + (\gamma_1 \varepsilon - \gamma_2)a^2 + (\gamma_2 \varepsilon - \gamma_1 \varepsilon)a - \gamma_2 \varepsilon$$

so Condition (iv) holds for sufficiently large a .

We also have

$$\gamma_1 y_2 b^2 - cb - \gamma_2(y_1 + y_2) = -((\gamma_2 + \gamma_3)b + \gamma_2)a^2 + \gamma_1 \varepsilon b - \gamma_2 \varepsilon b + \gamma_1 \varepsilon b - \gamma_2 \varepsilon.$$

Thus, if b satisfies $(\gamma_2 + \gamma_3)b + \gamma_2 > 0$, that is,

$$b > \frac{-\gamma_2}{\gamma_2 + \gamma_3} \quad (8)$$

then Condition (v) holds for sufficiently large a . Combining all these conditions on a and b , the matrix Q is admissible.

This is summarised in the following proposition.

Proposition 2.1 (Sufficient condition for admissibility ($N = 3$)). *Let Q denote the matrix defined by (6). If b is a real number satisfying*

$$b \geq \frac{\gamma_2}{\gamma_1} K \quad \text{and} \quad b > \frac{-\gamma_2}{\gamma_2 + \gamma_3}, \quad (9)$$

then there exists $a_0 \in \mathbb{R}$ such that, for every $a \geq a_0$, the matrix Q is admissible.

There are therefore infinitely many admissible matrices in dimension $N = 3$.

2.2 Invariant Domain

The following theorem gives a class of domains satisfying an invariance property for the multifactor process (3). This is the main result of [7]. We reproduce its proof to emphasise the difficult points and because it is adapted in the following sections to establish extensions. More generally, throughout Section 2.2, the stated results were proved in [7], but we reproduce each proof because their proof strategies will be reused in the extensions presented later.

Theorem 2.1 (Invariant domain). *Assume that $b, \sigma : \mathbb{R} \rightarrow \mathbb{R}$ are continuous with at most linear growth, that $\sigma(0) = 0$, and that $b(0) \geq 0$. Let $Q \in \mathcal{Q}$. Set $\mathbf{y}_0 = \mu \text{diag}(\boldsymbol{\rho})^{-1} \mathbf{1}$ with $\mu \geq 0$. Then, with $\mathcal{D} = Q^{-1} \mathbb{R}_+^N$, for every $\mathbf{Y}_0 \in \mathcal{D}$, there exists a weak solution $\mathbf{Y}$ of (3), starting from $\mathbf{Y}_0$, that remains in $\mathcal{D}$, in the sense that $\mathbf{Y}_t \in \mathcal{D}$ for every $t \geq 0$ almost surely.*

Proof. Set $b_\mu(z) = b(z) + \mu$. The idea of the proof is to make the change of variables $\mathbf{Z} = Q\mathbf{Y}$. Proving the theorem then amounts to showing that, for every initial condition $\mathbf{Z}_0 \in \mathbb{R}_+^N$, there exists a weak solution $\mathbf{Z}$ of

$$d\mathbf{Z}_t = -Q \text{diag}(\boldsymbol{\rho}) Q^{-1} \mathbf{Z}_t dt + b_\mu(\boldsymbol{\gamma}^\top Q^{-1} \mathbf{Z}_t) Q \mathbf{1} dt + \sigma(\boldsymbol{\gamma}^\top Q^{-1} \mathbf{Z}_t) Q \mathbf{1} dW_t \quad (10)$$

that remains in $\mathbb{R}_+^N$ over time.

Using Properties 2 and 3 of the admissibility of Q , the preceding equation can be rewritten as

$$d\mathbf{Z}_t = -Q \text{diag}(\boldsymbol{\rho}) Q^{-1} \mathbf{Z}_t dt + \bar{\gamma} b_\mu(Z_t^{(N)}) \mathbf{e}_N dt + \bar{\gamma} \sigma(Z_t^{(N)}) \mathbf{e}_N dW_t. \quad (11)$$

We apply Lemma A.1 and Corollary A.1 from [5] to show that equation (11) admits a weak solution that remains in $\mathbb{R}_+^N$. We therefore set

$$\tilde{b}(z) = -Q \text{diag}(\boldsymbol{\rho}) Q^{-1} z + \bar{\gamma} b_\mu(z_N) \mathbf{e}_N \quad \text{and} \quad \tilde{\sigma}(z) = \bar{\gamma} \sigma(z_N) \mathbf{e}_N$$

to rewrite equation (11) as

$$d\mathbf{Z}_t = \tilde{b}(\mathbf{Z}_t) dt + \tilde{\sigma}(\mathbf{Z}_t) dW_t$$

Let $\mathbf{z} \in \mathbb{R}_+^N$ be such that $z_i = 0$.
Then

$$\bar{\gamma} b_\mu(z_N) \mathbf{e}_i^\top \mathbf{e}_N = \begin{cases} 0 & \text{if } i \neq N \\ \bar{\gamma}(b(0) + \mu) & \text{otherwise.} \end{cases}$$

Thus, since $b(0)$ and μ are nonnegative, $\bar{\gamma} b_\mu(z_N) \mathbf{e}_i^\top \mathbf{e}_N \geq 0$, and hence

$$\begin{aligned} \mathbf{e}_i^\top \tilde{b}(\mathbf{z}) &\geq \mathbf{e}_i^\top Q \text{diag}(\boldsymbol{\rho}) Q^{-1} \mathbf{z} \\ &= - \sum_{j \neq i} [Q \text{diag}(\boldsymbol{\rho}) Q^{-1}]_{ij} z_j \quad \text{because } z_i = 0 \\ &\geq 0 \quad \text{by Property 4 of admissibility.} \end{aligned}$$

Moreover, $\mathbf{e}_i^\top \tilde{\sigma}(\mathbf{z}) = \bar{\gamma} \sigma(z_N) \mathbf{e}_i^\top \mathbf{e}_N$. Thus, if $i < N$, then $\mathbf{e}_i^\top \mathbf{e}_N = 0$, while if $i = N$, then $\sigma(z_N) = \sigma(0) = 0$. In all cases, $\mathbf{e}_i^\top \tilde{\sigma}(\mathbf{z}) = 0$.

Thus, for every $\mathbf{Z}_0 \in \mathbb{R}_+^N$, Lemma A.1 ensures the existence of a weak solution $\mathbf{Z}$ of (11), starting from $\mathbf{Z}_0$, such that $\mathbf{Z}_t \in \mathbb{R}_+^N$ for every $t \geq 0$. This completes the proof of Theorem 2.1. $\blacksquare$

Remark 2.2. When $N = 2$, one can show that the domain $\mathcal{D}$ is independent of the choice of $q \neq 0$ and is given by

$$\mathcal{D} = \{y \in \mathbb{R}_+^2 \mid \gamma^\top \mathbf{y} \geq 0, y_1 \geq y_2\}. \quad (12)$$

If $\mathbf{y}_0$ is chosen so that $y_0^{(i)} = Y_0^{(i)}$ and $\sum_{i=0}^N \gamma_i y_0^{(i)} = Y_0$, we have seen that a solution Y of the Volterra equation (1) can be written as $\sum_{i=1}^N \gamma_i Y_t^{(i)}$, where the $Y^{(i)}$ satisfy (3). It follows from Theorem 2.1 that the Volterra equation (1) satisfies an invariance property. It preserves the domain $\mathbb{R}_+$ in the sense that, if $Y_0 \geq 0$, then $Y_t \geq 0$ for every $t \geq 0$.

To obtain this result, we introduce two lemmas. The first states that elements $\mathbf{y}_0$ of the form imposed in Theorem 2.1 actually belong to $Q^{-1} \mathbb{R}_+^N$.

Lemma 2.1. If $Q \in \mathcal{Q}$ and $\mathbf{y}_0 \in \mathbb{R}^N$ can be written as

$$\mathbf{y}_0 = \mu \text{diag}(\boldsymbol{\rho})^{-1} \mathbf{1},$$

then $\mathbf{y}_0 \in Q^{-1} \mathbb{R}_+^N$.

Proof. We have

$$\begin{aligned} Q \mathbf{y}_0 &= \mu Q \text{diag}(\boldsymbol{\rho})^{-1} \mathbf{1} \\ &= \mu Q \text{diag}(\boldsymbol{\rho})^{-1} Q^{-1} Q \mathbf{1} \\ &= \mu \bar{\gamma} Q \text{diag}(\boldsymbol{\rho})^{-1} Q^{-1} \mathbf{e}_N \quad \text{by Property 3 of admissibility.} \end{aligned}$$

Now, the matrix $Q \text{diag}(\boldsymbol{\rho})^{-1} Q^{-1}$ is the inverse of $Q \text{diag}(\boldsymbol{\rho}) Q^{-1}$, which is an M-matrix (see Definition A.3) by Property 4 of admissibility and because its spectrum is $\{\rho_i, i \in \llbracket 1, N \rrbracket\}$. The entries of $Q \text{diag}(\boldsymbol{\rho})^{-1} Q^{-1}$ are therefore nonnegative by Proposition A.2, so $Q \mathbf{y}_0 \in \mathbb{R}_+^N$. $\blacksquare$

The second lemma allows the invariant domain $\mathcal{D}$ to be translated while preserving invariance.

Lemma 2.2 (Translation of the domain). Let $b, \sigma : \mathbb{R} \rightarrow \mathbb{R}$ be continuous functions with at most linear growth such that $\sigma(0) = 0$ and $b(0) \geq 0$. Let $Q \in \mathcal{Q}$, and let $\mathbf{y}_0 \in \mathbb{R}^N$ satisfy $\gamma^\top \mathbf{y}_0 \geq 0$. Set $\tilde{\mathbf{y}}_0 := \mu \text{diag}(\boldsymbol{\rho})^{-1} \mathbf{1}$, where $\mu \geq 0$ is chosen so that $\gamma^\top \tilde{\mathbf{y}}_0 = \gamma^\top \mathbf{y}_0$, and let

$\mathcal{D} := Q^{-1}\mathbb{R}_+^N + (\mathbf{y}_0 - \tilde{\mathbf{y}}_0)$. Then, for every $\mathbf{Y}_0 \in \mathcal{D}$, there exists a weak solution $\mathbf{Y}$ of (3), starting from $\mathbf{Y}_0$, that remains in $\mathcal{D}$.

Proof. Let $\mathbf{Y}_0 \in \mathcal{D}$. By Theorem 2.1, there exists a weak solution $\tilde{\mathbf{Y}}$ of

$$d\tilde{\mathbf{Y}}_t = -\text{diag}(\boldsymbol{\rho})(\tilde{\mathbf{Y}}_t - \tilde{\mathbf{y}}_0)dt + b(\boldsymbol{\gamma}^\top \tilde{\mathbf{Y}}_t)\mathbf{1}dt + \sigma(\boldsymbol{\gamma}^\top \tilde{\mathbf{Y}}_t)\mathbf{1}dW_t \quad (13)$$

starting from $\tilde{\mathbf{Y}}_0 = \mathbf{Y}_0 + \tilde{\mathbf{y}}_0 - \mathbf{y}_0 \in Q^{-1}\mathbb{R}_+^N$ and remaining in $Q^{-1}\mathbb{R}_+^N$. Set $\mathbf{R} = \mathbf{Y} + \mathbf{y}_0 - \tilde{\mathbf{y}}_0$. Then $\boldsymbol{\gamma}^\top \mathbf{R} = \boldsymbol{\gamma}^\top \tilde{\mathbf{Y}}_t$, and consequently

$$d\mathbf{R}_t = -\text{diag}(\boldsymbol{\rho})(\mathbf{R}_t - \mathbf{y}_0)dt + b(\boldsymbol{\gamma}^\top \mathbf{R}_t)\mathbf{1}dt + \sigma(\boldsymbol{\gamma}^\top \mathbf{R}_t)\mathbf{1}dW_t \quad (14)$$

with $\mathbf{R}_0 = \mathbf{Y}_0$. Moreover, the process $\mathbf{R}$ takes values in $\mathcal{D}$. $\blacksquare$

These two lemmas yield the following corollary.

Corollary 2.1 (Positivity of the Volterra equation). *Let $b, \sigma : \mathbb{R} \rightarrow \mathbb{R}$ be continuous functions with at most linear growth such that $\sigma(0) = 0$ and $b(0) \geq 0$. Then, for every $Y_0 \geq 0$, there exists a weak solution Y of the Volterra equation (1), starting from Y_0 , such that $Y_t \geq 0$ for every $t \geq 0$ almost surely.*

Proof. Let $Y_0 \geq 0$. We would like to apply Theorem 2.1 with $\mathbf{y}_0$, but $\mathbf{y}_0$ is not necessarily of the form $\mathbf{y}_0 = \mu \text{diag}(\boldsymbol{\rho})^{-1}\mathbf{1}$, so we cannot do so directly.

We therefore take $\tilde{\mathbf{y}}_0 = \mu \text{diag}(\boldsymbol{\rho})^{-1}\mathbf{1}$, with $\mu \geq 0$ given by

$$\mu = \frac{Y_0}{\boldsymbol{\gamma}^\top \text{diag}(\boldsymbol{\rho})^{-1}\mathbf{1}}, \quad (15)$$

so that $\boldsymbol{\gamma}^\top \tilde{\mathbf{y}}_0 = Y_0$.

Thus, for $Q \in \mathcal{Q}$, Lemma 2.1 gives $\tilde{\mathbf{y}}_0 \in Q^{-1}\mathbb{R}_+^N$. Hence $\mathbf{y}_0 = \tilde{\mathbf{y}}_0 + (\mathbf{y}_0 - \tilde{\mathbf{y}}_0) \in Q^{-1}\mathbb{R}_+^N + \mathbf{y}_0 - \tilde{\mathbf{y}}_0$, and Lemma 2.2 provides a weak solution $\mathbf{Y}$ of (3), starting from $\mathbf{y}_0$, that remains in $Q^{-1}\mathbb{R}_+^N + (\mathbf{y}_0 - \tilde{\mathbf{y}}_0)$.

Using variation of constants, we can write

$$\mathbf{Y}_t = \mathbf{y}_0 + \int_0^t \exp(-\text{diag}(\boldsymbol{\rho})(t-s))\mathbf{1} [b(\boldsymbol{\gamma}^\top \mathbf{Y}_s)ds + \sigma(\boldsymbol{\gamma}^\top \mathbf{Y}_s)dW_s].$$

Multiplying on the left by $\boldsymbol{\gamma}^\top$ and observing that $\boldsymbol{\gamma}^\top \mathbf{y}_0 = Y_0$ and $\boldsymbol{\gamma}^\top \exp(-\text{diag}(\boldsymbol{\rho})(t-s))\mathbf{1} = K(t-s)$ gives

$$Y_t := \boldsymbol{\gamma}^\top \mathbf{Y}_t = Y_0 + \int_0^t K(t-s)b(Y_s)ds + \int_0^t K(t-s)\sigma(Y_s)dW_s.$$

Thus, Y is a weak solution of the Volterra equation (1). It remains to check that Y takes nonnegative values, which indeed follows from

$$Y_t \in \boldsymbol{\gamma}^\top [\mathbf{y}_0 - \tilde{\mathbf{y}}_0 + Q^{-1}\mathbb{R}_+^N] = \mathbf{e}_N^\top Q Q^{-1}\mathbb{R}_+^N = \mathbb{R}_+$$

because $\boldsymbol{\gamma}^\top \mathbf{y}_0 = \boldsymbol{\gamma}^\top \tilde{\mathbf{y}}_0$ and by Property 2 of admissibility. Thus $Y_t \geq 0$ for every $t \geq 0$. $\blacksquare$

2.3 Adding a Constant

In this section, we study the effect on the domain of adding a constant to the solution of the Volterra equation (1). More precisely, instead of considering $Y_t = \sum_{i=1}^N \gamma_i Y_t^{(i)}$, we now

consider

$$Y_t = \alpha + \sum_{i=1}^N \gamma_i Y_t^{(i)}$$

where $\alpha \in \mathbb{R}$ and the $Y^{(i)}$ satisfy

$$\forall i \in \llbracket 1, N \rrbracket, \quad dY_t^{(i)} = -\rho_i \left[Y_t^{(i)} - y_0^{(i)} \right] dt + b(Y_t)dt + \sigma(Y_t)dW_t.$$

Setting $\mathbf{Y} = (Y^{(i)})_{i \in \llbracket 1, N \rrbracket}$, the preceding N equations can be written as a single N -dimensional equation:

$$d\mathbf{Y}_t = -\text{diag}(\boldsymbol{\rho}) [\mathbf{Y}_t - \mathbf{y}_0] dt + b(\boldsymbol{\gamma}^\top \mathbf{Y}_t + \alpha) \mathbf{1} dt + \sigma(\boldsymbol{\gamma}^\top \mathbf{Y}_t + \alpha) \mathbf{1} dW_t. \quad (16)$$

A natural question is the following:

Can we identify a class of domains $\mathcal{D} \subset \mathbb{R}^N$ such that, if $\mathbf{Y}_0 \in \mathcal{D}$, then there exists a weak solution $\mathbf{Y}$ of (16), starting from $\mathbf{Y}_0$, such that $\mathbf{Y}_t \in \mathcal{D}$ for every $t \geq 0$?

Under suitable assumptions, the answer is yes, as stated in the following proposition.

Proposition 2.2. *Let $\boldsymbol{\beta}$ satisfy $\boldsymbol{\gamma}^\top \boldsymbol{\beta} = \alpha$. Assume that $\mathbf{y}_0$ is of the form $\mathbf{y}_0 = -\boldsymbol{\beta} + \mu \text{diag}(\boldsymbol{\rho})^{-1} \mathbf{1}$, where $\mu \geq 0$. Set*

$$\mathcal{D} = -\boldsymbol{\beta} + Q^{-1} \mathbb{R}_+^N.$$

Then, for every $\mathbf{Y}_0 \in \mathcal{D}$, there exists a weak solution $\mathbf{Y}$ of (16), starting from $\mathbf{Y}_0$, such that $\mathbf{Y}_t \in \mathcal{D}$ for every $t \geq 0$.

Consequently, $\mathbf{Y}$ satisfies

$$\forall t \geq 0, \quad \alpha + \boldsymbol{\gamma}^\top \mathbf{Y}_t \geq 0.$$

Proof. Let $\boldsymbol{\beta}$ satisfy $\boldsymbol{\gamma}^\top \boldsymbol{\beta} = \alpha$. Setting $\mathbf{Z} = \mathbf{Y} + \boldsymbol{\beta}$, equation (16) becomes

$$d\mathbf{Z}_t = -\text{diag}(\boldsymbol{\rho}) [\mathbf{Z}_t - \mathbf{z}_0] dt + b(\boldsymbol{\gamma}^\top \mathbf{Z}_t) dt + \sigma(\boldsymbol{\gamma}^\top \mathbf{Z}_t) dW_t \quad (17)$$

with $\mathbf{z}_0 = \boldsymbol{\beta} + \mathbf{y}_0$.

By Theorem 2.1, if $\mathbf{z}_0 = \mu \text{diag}(\boldsymbol{\rho})^{-1} \mathbf{1}$ with $\mu \geq 0$, then, for every $\mathbf{Z}_0 \in Q^{-1} \mathbb{R}_+^N$, there exists a weak solution of (17), starting from $\mathbf{Z}_0$, that remains in $Q^{-1} \mathbb{R}_+^N$ over time.

It follows that, with $\mathbf{y}_0 = -\boldsymbol{\beta} + \mu \text{diag}(\boldsymbol{\rho})^{-1} \mathbf{1}$, if $\mathbf{Y}_0 \in \mathcal{D} := -\boldsymbol{\beta} + Q^{-1} \mathbb{R}_+^N$, then there exists a weak solution $\mathbf{Y}$ of (16), starting from $\mathbf{Y}_0$, such that $\mathbf{Y}_t \in \mathcal{D}$ for every $t \geq 0$ almost surely. $\blacksquare$

Proposition 2.2 yields the following corollary.

Corollary 2.2. *Assume that $\mathbf{y}_0 = 0$ and $\alpha \geq 0$. There exists a weak solution $\mathbf{Y}$ of (16), starting from 0, such that $\alpha + \boldsymbol{\gamma}^\top \mathbf{Y}_t \geq 0$ for every $t \geq 0$.*

Proof. Take $\boldsymbol{\beta} = \mu \text{diag}(\boldsymbol{\rho})^{-1} \mathbf{1}$, with $\mu \geq 0$ given by

$$\mu = \frac{\alpha}{\boldsymbol{\gamma}^\top \text{diag}(\boldsymbol{\rho})^{-1} \mathbf{1}}, \quad (18)$$

so that $\boldsymbol{\gamma}^\top \boldsymbol{\beta} = \alpha$.

Set $\mathcal{D} = -\boldsymbol{\beta} + Q^{-1} \mathbb{R}_+^N$. By Proposition 2.2, for every $\mathbf{Y}_0 \in \mathcal{D}$, there exists a weak solution $\mathbf{Y}$ of (16), starting from $\mathbf{Y}_0$, such that $\mathbf{Y}_t \in \mathcal{D}$ and $\alpha + \boldsymbol{\gamma}^\top \mathbf{Y}_t \geq 0$ for every $t \geq 0$.

Now, $0 \in \mathcal{D}$ because Lemma 2.1 gives $\boldsymbol{\beta} \in Q^{-1} \mathbb{R}_+^N$. Hence there exists a weak solution $\mathbf{Y}$ of (16), starting from 0, such that $\alpha + \boldsymbol{\gamma}^\top \mathbf{Y}_t \geq 0$ for every $t \geq 0$. $\blacksquare$

2.4 Compatibility of the Domains Across Dimensions

2.4.1 Motivation

Corollary 2.1 ensures that, under assumptions on b and σ , if the initial condition satisfies $Y_0 \geq 0$, then the Volterra equation

$$Y_t = Y_0 + \int_0^t K(t-s)b(Y_s)ds + \int_0^t K(t-s)\sigma(Y_s)dW_s$$

remains nonnegative when the kernel K is a Prony series, that is, of the form

$$K(t) = \sum_{i=1}^N \gamma_i e^{-\rho_i t} \quad \text{with } 0 < \rho_1 < \dots < \rho_N \text{ and } \forall i \in \llbracket 1, N \rrbracket, \gamma_i > 0.$$

If we choose

$$\forall i \in \llbracket 1, N \rrbracket \quad \gamma_i = \frac{1}{N} \quad \text{and} \quad \rho_i = \frac{i}{N},$$

then the kernel K can be written as $K_N(t) = \frac{1}{N} \sum_{i=1}^N e^{-\frac{i}{N}t}$. By the theorem on Riemann sums, since the function $x \mapsto e^{-xt}$ is continuous on $[0, 1]$, we have

$$K_N(t) \xrightarrow[n \rightarrow \infty]{} K_\infty(t) := \int_0^1 e^{-xt} dx.$$

The kernel K_∞ can be written as

$$K_\infty(t) = \begin{cases} \frac{1}{t}(e^{-t} - 1) & \text{if } t \neq 0 \\ 1 & \text{otherwise.} \end{cases}$$

It is therefore natural to seek an analogue of Corollary 2.1 when the kernel has this form. We thus attempt to pass to the limit $N \rightarrow \infty$ in Theorem 2.1. To this end, the following sections investigate whether, upon adding a weight γ_0 , the domain $\tilde{\mathcal{D}}$ in dimension $N+1$ can be made compatible with the domain $\mathcal{D}$ in dimension N , that is, whether $\tilde{\mathcal{D}}$ can be expressed in terms of $\mathcal{D}$. We therefore examine whether an admissible matrix $\tilde{Q}$ in dimension $N+1$ can be constructed from a matrix Q in dimension N .

2.4.2 Compatibility of Admissibility

Under suitable assumptions, Theorem 2.1 gives a class of domains $\mathcal{D}$ such that, for every $\mathbf{Y}_0 \in \mathcal{D}$, the N -dimensional equation (3) admits a weak solution starting from $\mathbf{Y}_0$ and remaining in $\mathcal{D}$. We now ask whether such a domain $\mathcal{D}$ can be used to construct a domain $\tilde{\mathcal{D}} \subset \mathbb{R}^{N+1}$ such that, for every $\mathbf{Y}_0 \in \tilde{\mathcal{D}}$, the $(N+1)$ -dimensional equation (3) admits a weak solution $\mathbf{Y}$ starting from $\mathbf{Y}_0$ and remaining in $\tilde{\mathcal{D}}$.

Let Q be an admissible matrix in dimension N for the parameter sets $(\gamma_1, \dots, \gamma_N)$ and $(\rho_1, \dots, \rho_N)$. Let $\gamma_0 > 0$ and $0 < \rho_0 < \rho_1$. We seek to construct from Q an admissible matrix $\tilde{Q}$ in dimension $N+1$ for the parameter sets $(\gamma_0, \gamma_1, \dots, \gamma_N)$ and $(\rho_0, \rho_1, \dots, \rho_N)$.

For $\tilde{Q}$ to have any chance of being admissible, it must satisfy several properties. First, it is natural to place Q in the lower-right block of $\tilde{Q}$ and impose $\tilde{Q}_{N+1,1} = \gamma_0$, since Property 2 of admissibility requires the last row of $\tilde{Q}$ to be $(\gamma_0 \gamma_1 \dots \gamma_N)$. Next, the entries in the first row of $\tilde{Q}$ must sum to zero by Property 3 of admissibility. We therefore choose this row to be $(-\sum_{k=1}^N q_k \quad q_1 \dots q_N)$. The remaining entries are set to 0, since every row of $\tilde{Q}$ except the last must sum to zero by Property 3. Altogether, we take $\tilde{Q}$ of the form

$$\tilde{Q} = \left(\begin{array}{c|ccc} -\sum_{k=1}^N q_k & q_1 & \dots & q_N \\ 0 & & & \\ \vdots & & & \\ 0 & & & \\ \gamma_0 & & & \end{array} \middle| \begin{array}{c} Q \\ \\ \\ \\ \end{array} \right). \quad (19)$$

We now determine the conditions under which $\tilde{Q}$ is admissible in this form. Properties 2 and 3 of admissibility hold automatically for $\tilde{Q}$, since its form was chosen precisely for that purpose. We therefore focus on Properties 1 and 4.

Property 1. We choose the q_i so as to make $\tilde{Q}$ invertible and express its inverse explicitly. We determine the conditions under which the matrix

$$\tilde{R} = \left(\begin{array}{c|c} \alpha & \mathbf{u}^\top \\ \mathbf{v} & M \end{array} \right)$$

is the inverse of $\tilde{Q}$. A block calculation shows that

$$\tilde{R}\tilde{Q} = \left(\begin{array}{c|c} \beta & L \\ C & A \end{array} \right),$$

where

$$\begin{cases} \beta = -\alpha \sum_{k=1}^N q_k + u_N \gamma_0 \\ L = \alpha(q_1 \dots q_N) + \mathbf{u}^\top Q \\ C = -\mathbf{v} \sum_{k=1}^N q_k + M \begin{pmatrix} 0 \\ \vdots \\ 0 \\ \gamma_0 \end{pmatrix} \\ A = \mathbf{v}(q_1 \dots q_N) + MQ. \end{cases}$$

The required conditions are therefore $\beta = 1$, $L = 0$, $C = 0$ and $A = I_N$. These conditions are equivalent to

$$\begin{cases} \alpha = \frac{-1}{\sum_{k=1}^N q_k + \gamma_0 \sum_{k=1}^N q_k [Q^{-1}]_{kN}} \\ \mathbf{u}^\top = -\alpha(q_1 \dots q_N)Q^{-1} \\ \forall i \in \llbracket 1, N \rrbracket, \quad v_i = \frac{\gamma_0 [Q^{-1}]_{iN}}{\sum_{k=1}^N q_k + \gamma_0 \sum_{k=1}^N q_k [Q^{-1}]_{kN}} \\ M = -\mathbf{v}(q_1 \dots q_N)Q^{-1} + Q^{-1}. \end{cases} \quad (20)$$

Under these conditions, the matrix $\tilde{Q}$ is invertible, with inverse $\tilde{R}$. At this stage, Properties 1, 2 and 3 hold. It remains to address Property 4.

Property 4. A block calculation shows that

$$\tilde{Q} \text{diag}(\rho_0, \rho_1, \dots, \rho_N) \tilde{Q}^{-1} = \left(\begin{array}{c|c} \gamma & L' \\ C' & A' \end{array} \right)$$

where γ , L' , C' and A' are given by

$$\begin{cases} \gamma = -\rho_0 \alpha \sum_{k=1}^N q_k + (q_1 \dots q_N) \text{diag}(\boldsymbol{\rho}) \mathbf{v} \\ L' = -\rho_0 \left(\sum_{k=1}^N q_k \right) \mathbf{u}^\top + (q_1 \dots q_N) \text{diag}(\boldsymbol{\rho}) M \\ C' = \rho_0 \alpha \begin{pmatrix} 0 \\ \vdots \\ 0 \\ \gamma_0 \end{pmatrix} + Q \text{diag}(\boldsymbol{\rho}) \mathbf{v} \\ A' = \rho_0 \begin{pmatrix} 0 \\ \vdots \\ 0 \\ \gamma_0 \end{pmatrix} \mathbf{u}^\top + Q \text{diag}(\boldsymbol{\rho}) M. \end{cases}$$

We must therefore impose

$$\begin{cases} [L']_i \leq 0, [C']_i \leq 0 & \text{for every } i \in \llbracket 1, N \rrbracket, \\ [A']_{ij} \leq 0 & \text{for all } i \neq j \in \llbracket 1, N \rrbracket. \end{cases} \quad (21)$$

The preceding discussion is summarised in the following proposition.

Proposition 2.3 (Compatibility of admissibility). *Let $Q \in \mathcal{Q}$. Let $\gamma_0 > 0$ and $\rho_0 \in]0, \rho_1[$. Furthermore, let $(q_k)_{k \in \llbracket 1, N \rrbracket}$ satisfy conditions (20) and (21). Then the matrix $\tilde{Q}$ defined in (19) is admissible.*

2.4.3 Compatibility of the Domains

Under the conditions of Proposition 2.3, the matrix $\tilde{Q}$ is admissible. Hence, by Theorem 2.1, with $\tilde{\mathbf{y}}_0 = \mu \text{diag}(\rho_0, \rho_1, \dots, \rho_N)^{-1} \mathbf{1}_{N+1}$, with b and σ satisfying the assumptions of Theorem 2.1, and with $\tilde{\mathcal{D}} = \tilde{Q}^{-1} \mathbb{R}_+^{N+1}$, for every $\tilde{\mathbf{Y}}_0 \in \tilde{\mathcal{D}}$, the $(N+1)$ -dimensional equation (3) admits a weak solution $\tilde{\mathbf{Y}}$ starting from $\tilde{\mathbf{Y}}_0$ and remaining in $\tilde{\mathcal{D}}$. The domain $\tilde{\mathcal{D}}$ can be written as

$$\tilde{\mathcal{D}} = \begin{pmatrix} \alpha \\ \mathbf{v} \end{pmatrix} \mathbb{R}_+ + \mathcal{E} \quad (22)$$

with

$$\mathcal{E} = \left\{ \begin{pmatrix} -\alpha(q_1 \dots q_N) \mathbf{y} \\ -\mathbf{v}(q_1 \dots q_N) \mathbf{y} \end{pmatrix} \mid \mathbf{y} \in \mathcal{D} \right\}.$$

The following proposition summarises the result just obtained.

Proposition 2.4 (Compatibility of the domains under an increase in dimension). *Let $Q \in \mathcal{Q}$ in dimension N , and let $s \neq 0$. Let $0 < \rho_1 < \dots < \rho_N$ and $\gamma_1, \dots, \gamma_N > 0$. Let $b, \sigma : \mathbb{R} \rightarrow \mathbb{R}$ be continuous functions with at most linear growth such that $\sigma(0) = 0$ and $b(0) \geq 0$. Let $\rho_0 \in]0, \rho_1[$, $\gamma_0 > 0$, and let $(q_i)_{i \in \llbracket 1, N \rrbracket}$ satisfy properties (20) and (21). Set*

$$\tilde{\mathbf{y}}_0 = \mu \text{diag}(\rho_0, \rho_1, \dots, \rho_N)^{-1} \mathbf{1}_{N+1} \quad \text{with } \mu \geq 0.$$

Then, defining $\tilde{\mathcal{D}}$ as in (22), for every $\tilde{\mathbf{Y}}_0 \in \tilde{\mathcal{D}}$, there exists a weak solution of the $(N+1)$ -dimensional equation (3), starting from $\tilde{\mathbf{Y}}_0$, that remains in $\tilde{\mathcal{D}}$.

2.4.4 Cases $N \in \{1, 2, 3\}$

The conditions for $\tilde{Q}$ to be admissible in Proposition 2.3 are abstract. It is not known whether one can find $(q_1, \dots, q_N)$ satisfying them in general. In this section, we consider the cases $N \in \{1, 2\}$.

Case $N = 1$. In this case $Q = (\gamma_1)$, and $\tilde{Q}$ is of the form

$$\tilde{Q} = \begin{pmatrix} q_1 & -q_1 \\ \gamma_0 & \gamma_1 \end{pmatrix}.$$

which is admissible if and only if $q_1 > 0$, according to [7]. The associated domain $\tilde{D}$ does not depend on q_1 and is given by (12).

Case $N = 2$. This time, we start from

$$Q = \begin{pmatrix} q & -q \\ \gamma_1 & \gamma_2 \end{pmatrix}$$

with $q > 0$.

The matrix $\tilde{Q}$ can be written as

$$\tilde{Q} = \begin{pmatrix} -q_1 - q_2 & q_1 & q_2 \\ 0 & q & -q \\ \gamma_0 & \gamma_1 & \gamma_2 \end{pmatrix}.$$

We have

$$\det(\tilde{Q}) = -q(q_1 + q_2)(\gamma_1 + \gamma_2)$$

so, taking $q \neq 0$, $\tilde{Q}$ is invertible. We must therefore impose $q_1 + q_2 \neq 0$.

Set

$$T = \tilde{Q} \text{diag}(\rho_0, \rho_1, \rho_2) \tilde{Q}^{-1}.$$

A calculation shows that

$$T_{2,1} = \frac{q\gamma_0}{q_1 + q_2}(\rho_1 - \rho_2), \quad T_{3,1} = \frac{\gamma_0\gamma_2(\rho_2 - \rho_0) + \gamma_0\gamma_1(\rho_1 - \rho_0)}{(q_1 + q_2)(\gamma_0 + \gamma_1 + \gamma_2)}.$$

Thus, for $\tilde{Q}$ to be admissible, we require both $q_1 + q_2 \geq 0$ and $q_1 + q_2 \leq 0$. This is impossible because $q_1 + q_2 \neq 0$. Hence the conditions of Proposition 2.3 are never satisfied.

Remark 2.3. *Except in the case $N = 1$, it appears that the conditions of Proposition 2.3 are never satisfied. We were unable to prove this in general and established it only when $N = 2$. In the case $N = 3$, we obtained relations suggesting that the conditions of Proposition 2.3 were unlikely to be satisfied. These calculations are omitted to avoid burdening the text unnecessarily.*

2.4.5 Using the Matrix Given in [7]

Another approach to obtaining a compatibility result for the domains is to start from the admissible matrix (4) given in [7]. This admissible matrix Q_N in dimension N satisfies the recurrence relation

$$Q_{N+1} = \left(\begin{array}{ccc|c} & & & 0 \\ & & & \vdots \\ & & & 0 \\ & & & -\sum_{k=1}^N \gamma_k \\ \hline \gamma_1 & \cdots & \gamma_N & \gamma_{N+1} \end{array} \right).$$

This gives some hope that the associated domains are compatible. However, according to [7], the inverse of Q_N is the matrix $R_N = (r_{i,j})_{i,j \in \llbracket 1, N \rrbracket}$ defined by

$$\begin{cases} r_{i,N} = \frac{1}{\gamma} & \text{for } i \in \llbracket 1, N \rrbracket \\ r_{i,j} = \frac{\gamma_{j+1}}{(\sum_{l=1}^j \gamma_l)(\sum_{l=1}^{j+1} \gamma_l)} & \text{for } i \leq j < N \\ r_{i+1,i} = \frac{-1}{\sum_{l=1}^{i+1} \gamma_l} & \text{for } i \in \llbracket 1, N-1 \rrbracket \\ r_{i,j} = 0 & \text{for } i \geq j+2. \end{cases}$$

It is clear that expressing R_{N+1} in terms of R_N is difficult. Consequently, the domain $R_{N+1}\mathbb{R}_+^{N+1}$ cannot be expressed in terms of the domain $R_N\mathbb{R}_+^N$.

Our attempt to extend Theorem 2.1 to kernels other than Prony series through a limiting argument therefore failed, but it provided further insight into the structure of admissible matrices.

2.5 Multidimensional Volterra Process

In [7], the Volterra process is one-dimensional, and Theorem 2.1 establishes an invariance result for the associated multifactor process in this setting.

In this section, we extend the results of [7] to dimension d . We start from the solution Y_t of a d -dimensional Volterra equation

$$Y_t = Y_0 + \int_0^t K(t-s)b(Y_s)ds + \int_0^t K(t-s)\sigma(Y_s)dW_s \quad (23)$$

where $b : \mathbb{R}^d \rightarrow \mathbb{R}^d$, $\sigma : \mathbb{R}^d \rightarrow \mathbb{R}^{d \times d}$, and $(W_t)_{t \geq 0}$ is a d -dimensional Brownian motion. Write $b = (b_1, \dots, b_d)^\top$, where $b_i : \mathbb{R}^d \rightarrow \mathbb{R}$, and $\sigma = (\sigma_1, \dots, \sigma_d)^\top$, where $\sigma_i : \mathbb{R}^d \rightarrow \mathbb{R}^d$ is the i -th row of the matrix σ .

We can still write Y_t as $Y_t = \sum_{i=1}^N \gamma_i Y_t^{(i)}$, where the $Y_t^{(i)} = (Y_{1t}^{(i)}, \dots, Y_{dt}^{(i)})$ are $\mathbb{R}^d$ -valued processes satisfying

$$dY_t^{(i)} = -\rho_i \left(Y_t^{(i)} - y_0^{(i)} \right) dt + b(Y_t) + \sigma(Y_t)dW_t \quad \forall i \in \llbracket 1, d \rrbracket \quad (24)$$

with $y_0^{(i)} \in \mathbb{R}^d$.

Remark 2.4. The Volterra process Y_t takes values in $\mathbb{R}^d$. Although Y_t is a vector, it will not be written in bold in this section. By analogy with the notation in the preceding sections, where $d = 1$, only vectors in $\mathbb{R}^{N^d}$ will be written in bold. We write vectors $\mathbf{z} \in \mathbb{R}^{N^d}$ as $\mathbf{z} = (z_1, \dots, z_d)$, where, for every $i \in \llbracket 1, d \rrbracket$, z_i is a vector in $\mathbb{R}^N$ with entries denoted by $z_i^1, \dots, z_i^N$.

For $i \in \llbracket 1, d \rrbracket$, let $Y_{it} = (Y_{it}^{(1)}, \dots, Y_{it}^{(N)})$ denote the vector of factors associated with dimension i . Set $\mathbf{Y}_t = (Y_{1t}, \dots, Y_{dt})$. This is a vector in $\mathbb{R}^{N^d}$ whose entries are, successively, the N factors

associated with dimension 1, then the N factors associated with dimension 2, and so on up to the N factors associated with dimension d (see Remark 2.4). The N equations (24) in dimension d can be written as the Nd -dimensional equation

$$d\mathbf{Y}_t = \tilde{b}(\mathbf{Y}_t)dt + \tilde{\sigma}(\mathbf{Y}_t)dW_t \quad (25)$$

where $\tilde{b} = (\tilde{b}_1, \dots, \tilde{b}_d)^\top : \mathbb{R}^{Nd} \longrightarrow \mathbb{R}^{Nd}$ is such that

$$\forall i \in \llbracket 1, d \rrbracket, \forall x \in \mathbb{R}^{Nd} \quad \tilde{b}_i(x) = \begin{pmatrix} -\rho_1(x_i^1 - y_{i0}^{(1)}) + b_i(\Gamma x) \\ \vdots \\ -\rho_N(x_i^N - y_{i0}^{(N)}) + b_i(\Gamma x) \end{pmatrix} \in \mathbb{R}^N$$

and $\tilde{\sigma} = (\tilde{\sigma}_1, \dots, \tilde{\sigma}_d)^\top : \mathbb{R}^{Nd} \longrightarrow \mathbb{R}^{Nd \times d}$ is such that

$$\forall x \in \mathbb{R}^{Nd}, \quad \tilde{\sigma}_i(x) = \begin{pmatrix} \sigma_i(\Gamma x) \\ \vdots \\ \sigma_i(\Gamma x) \end{pmatrix} \in \mathbb{R}^{N \times d}$$

with $\Gamma = \text{diag}(\gamma^\top, \dots, \gamma^\top) \in \mathbb{R}^{d \times Nd}$.

The following theorem generalises Theorem 2.1 to dimension d .

Theorem 2.2 (Invariant domain).

$$\begin{cases} b_i(x_1, \dots, x_{i-1}, 0, x_{i+1}, \dots, x_d) \geq 0 & \text{for every } i \in \llbracket 1, d \rrbracket, \\ \sigma_{ij}(x_1, \dots, x_{i-1}, 0, x_{i+1}, \dots, x_d) = 0 & \text{for all } i, j \in \llbracket 1, d \rrbracket. \end{cases} \quad (26)$$

Let $Q \in \mathcal{Q}$. Let $\mathbf{y}_0 = (y_{01}, \dots, y_{0d}) \in \mathbb{R}^{Nd}$, where, for $i \in \llbracket 1, d \rrbracket$, y_i is of the form $y_i = \mu_i \text{diag}(\boldsymbol{\rho})^{-1} \mathbf{1}$ with $\mu_i \geq 0$. Then, with $\mathcal{D} = (Q^{-1} \mathbb{R}_+^N)^d$, for every $\mathbf{Y}_0 \in \mathcal{D}$, there exists a weak solution $\mathbf{Y}$ of (25), starting from $\mathbf{Y}_0$, such that $\mathbf{Y}_t \in \mathcal{D}$ for every $t \geq 0$ almost surely.

Proof. We adapt the proof of 2.1. Let $\tilde{Q} = \text{diag}(Q, \dots, Q) \in \mathbb{R}^{Nd \times Nd}$. The matrix $\tilde{Q}$ is block diagonal, with d diagonal blocks all equal to Q . Set $\mathbf{Z}_t = \tilde{Q} \mathbf{Y}_t$. By (25),

$$d\mathbf{Z}_t = B(\mathbf{Z}_t)dt + V(\mathbf{Z}_t)dW_t \quad (27)$$

where $B(z) = \tilde{Q} \tilde{b}(\tilde{Q}^{-1}z)$ and $V(z) = \tilde{Q} \tilde{\sigma}(\tilde{Q}^{-1}z)$. To prove the result, it is enough to show that, for every $\mathbf{Z}_0 \in \mathbb{R}_+^{Nd}$, there exists a weak solution of (27) such that $\mathbf{Z}_t \in \mathbb{R}_+^{Nd}$ for every $t \geq 0$ almost surely. We apply Lemma A.1.

First, under the assumptions on σ , clearly $V(0) = 0$. Moreover, by Property 3 of admissibility,

$$B(0) = \begin{pmatrix} b_1(0)Q\mathbf{1} \\ \vdots \\ b_d(0)Q\mathbf{1} \end{pmatrix} = \bar{\gamma} \begin{pmatrix} b_1(0)\mathbf{e}_N \\ \vdots \\ b_d(0)\mathbf{e}_N \end{pmatrix},$$

whose entries are nonnegative by the assumption on b .

Let $\mathbf{z} \in \mathbb{R}_+^{Nd}$ be such that $z_i = 0$. Let $d_i \in \llbracket 1, d \rrbracket$ denote the dimension index such that

$i \in \llbracket Nd_i - N + 1, Nd_i \rrbracket$. A matrix calculation shows that

$$\begin{aligned} \mathbf{e}_i^\top B(z) &= -\mathbf{e}_i^\top \begin{pmatrix} Q \text{diag}(\boldsymbol{\rho}) Q^{-1} z_1 \\ \vdots \\ Q \text{diag}(\boldsymbol{\rho}) Q^{-1} z_d \end{pmatrix} + \mathbf{e}_i^\top \begin{pmatrix} Q \text{diag}(\boldsymbol{\rho}) y_{01} \\ \vdots \\ Q \text{diag}(\boldsymbol{\rho}) y_{0d} \end{pmatrix} + \mathbf{e}_i^\top \begin{pmatrix} b_1(\Gamma \tilde{Q}^{-1} z) Q \mathbf{1} \\ \vdots \\ b_d(\Gamma \tilde{Q}^{-1} z) Q \mathbf{1} \end{pmatrix} \\ &= v_1 + v_2 + v_3. \end{aligned}$$

For the first term, Property 4 of admissibility gives

$$v_1 = - \sum_{j \neq i} (Q \text{diag}(\boldsymbol{\rho}) Q^{-1})_{ij} z_{d_i j} \geq 0.$$

For the second term, since $y_{0d_i} = \mu_{d_i} \text{diag}(\boldsymbol{\rho})^{-1} \mathbf{1}$, Property 3 of admissibility gives

$$Q \text{diag}(\boldsymbol{\rho}) y_{0d_i} = \mu_{d_i} \bar{\gamma} \mathbf{e}_N.$$

Thus the entries of the vector to the right of $\mathbf{e}_i^\top$ in v_2 are nonnegative. Finally, for the last term,

$$v_3 = \bar{\gamma} \mathbf{e}_i^\top \begin{pmatrix} b_1(\Gamma \tilde{Q}^{-1} z) \mathbf{e}_N \\ \vdots \\ b_d(\Gamma \tilde{Q}^{-1} z) \mathbf{e}_N \end{pmatrix}$$

by Property 3 of admissibility. Observe that

$$\Gamma \tilde{Q}^{-1} z = \begin{pmatrix} \boldsymbol{\gamma}^\top Q^{-1} z_1 \\ \vdots \\ \boldsymbol{\gamma}^\top Q^{-1} z_d \end{pmatrix} = \begin{pmatrix} \mathbf{e}_N^\top z_1 \\ \vdots \\ \mathbf{e}_N^\top z_d \end{pmatrix} = \begin{pmatrix} z_1^N \\ \vdots \\ z_d^N \end{pmatrix}$$

by Property 2 of admissibility. Thus,

$$v_3 = \bar{\gamma} \mathbf{e}_i^\top \begin{pmatrix} b_1(z_1^N, \dots, z_d^N) \\ \vdots \\ b_d(z_1^N, \dots, z_d^N) \end{pmatrix}$$

Therefore, if $i \neq Nd_i$, then $v_3 = 0$, while if $i = Nd_i$, then $v_3 = \bar{\gamma} b_{d_i}(z_1^N, \dots, z_{d_i-1}^N, 0, z_{d_i+1}^N, \dots, z_d^N)$ which is nonnegative by the assumption on b .

Moreover,

$$\begin{aligned} \mathbf{e}_i^\top V(z) &= \mathbf{e}_i^\top \tilde{Q} \tilde{\sigma}(\tilde{Q}^{-1} z) \\ &= \mathbf{e}_i^\top \tilde{Q} \begin{pmatrix} \mathbf{1} \sigma_1(\Gamma \tilde{Q}^{-1} z) \\ \vdots \\ \mathbf{1} \sigma_d(\Gamma \tilde{Q}^{-1} z) \end{pmatrix} \\ &= \mathbf{e}_i^\top \begin{pmatrix} Q \mathbf{1} \sigma_1(\Gamma \tilde{Q}^{-1} z) \\ \vdots \\ Q \mathbf{1} \sigma_d(\Gamma \tilde{Q}^{-1} z) \end{pmatrix} \\ &= \bar{\gamma} \mathbf{e}_i^\top \begin{pmatrix} \sigma_1(z_1^N, \dots, z_d^N) \mathbf{e}_N \\ \vdots \\ \sigma_d(z_1^N, \dots, z_d^N) \mathbf{e}_N \end{pmatrix} \quad \text{by Property 3 of admissibility.} \end{aligned}$$

If $i \neq Nd_i$, it is therefore clear that $\mathbf{e}_i^\top V(z) = 0$, while if $i = Nd_i$, then

$$\mathbf{e}_i^\top V(z) = \bar{\gamma} \sigma_{d_i}(z_1^N, \dots, z_{d_i-1}^N, 0, z_{d_i+1}^N, \dots, z_d^N) = 0$$

by the assumption on σ .

Thus, by Lemma A.1, for every $\mathbf{Z}_0 \in \mathbb{R}_+^{Nd}$, there exists a weak solution $\mathbf{Z}$ of (27), starting from $\mathbf{Z}_0$, such that $\mathbf{Z}_t \in \mathbb{R}_+^{Nd}$ for every $t \geq 0$. This completes the proof. $\blacksquare$

The following lemma is the analogue of Lemma 2.2.

Lemma 2.3 (Translation of the domain in dimension d). *Let $b : \mathbb{R}^d \rightarrow \mathbb{R}^d$ and $\sigma : \mathbb{R}^d \rightarrow \mathbb{R}^{d \times d}$ be continuous functions satisfying conditions (26). Let $\mathbf{y}_0 \in \mathbb{R}^{Nd}$ be such that $\Gamma \mathbf{y}_0 \in \mathbb{R}_+^d$. Let $\tilde{\mathbf{y}}_0 = (\tilde{y}_{01}, \dots, \tilde{y}_{0d})$, where, for every $i \in \llbracket 1, d \rrbracket$,*

$$\tilde{y}_{0i} = \mu_i \operatorname{diag}(\boldsymbol{\rho})^{-1} \mathbf{1},$$

with $\mu_i \geq 0$ chosen so that $\Gamma \mathbf{y}_0 = \Gamma \tilde{\mathbf{y}}_0$, and set $\mathcal{D} = \tilde{Q} \mathbb{R}_+^{Nd} + (\mathbf{y}_0 - \tilde{\mathbf{y}}_0)$. Then, for every $\mathbf{Y}_0 \in \mathcal{D}$, there exists a weak solution $\mathbf{Y}$ of (25), starting from $\mathbf{Y}_0$, such that $\mathbf{Y}_t \in \mathbb{R}_+^{Nd}$ for every $t \geq 0$.

Proof. Let $\mathbf{Y}_0 \in \mathcal{D}$. Set

$$\forall i \in \llbracket 1, d \rrbracket, \quad \mu_i = \frac{\boldsymbol{\gamma}^\top \mathbf{y}_{0i}}{\boldsymbol{\gamma}^\top \operatorname{diag}(\boldsymbol{\rho})^{-1} \mathbf{1}}.$$

Define $\tilde{\mathbf{Y}}_0 = \mathbf{Y}_0 + \tilde{\mathbf{y}}_0 - \mathbf{y}_0$. Then $\tilde{\mathbf{Y}}_0 \in (Q^{-1} \mathbb{R}_+^N)^d$. Thus, by Theorem 2.2, there exists a weak solution $\tilde{\mathbf{Y}}$ of (25) with $\tilde{\mathbf{y}}_0$, starting from $\tilde{\mathbf{Y}}_0$, that remains in $(Q^{-1} \mathbb{R}_+^N)^d$. Set $\mathbf{R} := \tilde{\mathbf{Y}} + \mathbf{y}_0 - \tilde{\mathbf{y}}_0$. The process $\mathbf{R}$ satisfies (25) with $\mathbf{y}_0$. Moreover, $\mathbf{R}_t \in \mathcal{D}$ for every $t \geq 0$ almost surely. This completes the proof. $\blacksquare$

Lemma 2.3 yields the analogue of Corollary 2.1 in dimension d .

Corollary 2.3. *Let $b : \mathbb{R}^d \rightarrow \mathbb{R}^d$ and $\sigma : \mathbb{R}^d \rightarrow \mathbb{R}^{d \times d}$ be continuous functions satisfying conditions (26). Let $Y_0 \in \mathbb{R}_+^d$. Then there exists a weak solution Y of (23), starting from Y_0 , such that $\mathbf{Y}_t \in \mathbb{R}_+^d$ for every $t \geq 0$.*

Proof. Let $Y_0 \in \mathbb{R}_+^d$. Let $\mathbf{y}_0 \in \mathbb{R}^{Nd}$ satisfy $\Gamma \mathbf{y}_0 = Y_0$. Also let $\tilde{\mathbf{y}}_0 = (\tilde{y}_{01}, \dots, \tilde{y}_{0d})$, with $\tilde{y}_{0i} = \mu_i \operatorname{diag}(\boldsymbol{\rho})^{-1} \mathbf{1}$ and $\mu_i \geq 0$ chosen so that $\Gamma \tilde{\mathbf{y}}_0 = \Gamma \mathbf{y}_0$. By Lemma 2.1, $\tilde{y}_{0i} \in Q^{-1} \mathbb{R}_+^N$, and hence $\tilde{\mathbf{y}}_0 \in (Q^{-1} \mathbb{R}_+^N)^d$. Thus,

$$\tilde{\mathbf{y}}_0 = \mathbf{y}_0 + (\tilde{\mathbf{y}}_0 - \mathbf{y}_0) \in (Q^{-1} \mathbb{R}_+^N)^d + \tilde{\mathbf{y}}_0 - \mathbf{y}_0.$$

Therefore, by Lemma 2.3, there exists a weak solution $\mathbf{Y}$ of equation (25), starting from $\mathbf{y}_0$, such that $\mathbf{Y}_t \in (Q^{-1} \mathbb{R}_+^N)^d + \tilde{\mathbf{y}}_0 - \mathbf{y}_0$ for every $t \geq 0$ almost surely.

Set $Y_t = \Gamma \mathbf{Y}_t$. Variation of constants gives

$$\mathbf{Y}_t = \mathbf{y}_0 + \int_0^t \operatorname{diag}(\exp(-\operatorname{diag}(\boldsymbol{\rho})(t-s)), \dots, \exp(-\operatorname{diag}(\boldsymbol{\rho})(t-s))) \begin{pmatrix} \mathbf{1} \\ \vdots \\ \mathbf{1} \end{pmatrix} (b(Y_s)ds + \sigma(Y_s)dW_s).$$

Multiplying on the left by Γ yields

$$Y_t = Y_0 + \int_0^t K(t-s) (b(Y_s)ds + \sigma(Y_s)dW_s)$$

Finally, since $\mathbf{Y}_t \in (Q^{-1}\mathbb{R}_+^N)^d + \tilde{\mathbf{y}}_0 - \mathbf{y}_0$ and $\Gamma\tilde{\mathbf{y}}_0 = \Gamma\mathbf{y}_0$, Property 2 of admissibility readily shows that every entry of $\mathbf{Y}_t$ is nonnegative for every $t \geq 0$. This completes the proof. ■

3 Applications

3.1 Simulation Scheme

A simulation scheme is a function ψ used to generate a discretised $\mathbb{R}^N$ -valued trajectory $\mathbf{X}$ of the solution of a stochastic differential equation.

More precisely, let $(Z_l)_{l \in \mathbb{N}^*}$ be a sequence of i.i.d. random variables with the law of an $\mathbb{R}^{d_Z}$ -valued random variable Z , and let $T > 0$ and $N \in \mathbb{N}^*$. A simulation scheme is a function

$$\varphi : \mathbb{R}^N \times \mathbb{R}_+ \times \mathbb{R}^{d_Z} \longrightarrow \mathbb{R}^N$$

that generates the discretised trajectory returned by Algorithm 1.

Algorithm 1: Simulation scheme

Data: $\mathbf{x}_0, T, N, \psi$

Result: Simulated trajectory $\mathbf{X}_{t_k}$ for $k = 0, \dots, N$

```

1  $\mathbf{X}_0 = \mathbf{x}_0$ 
2  $\Delta T \leftarrow T/N$ 
3 for  $k = 1$  to  $N$  do
4   Draw  $Z_k \sim \mathcal{N}(0, 1)$ 
5    $\mathbf{X}_{k\Delta T} \leftarrow \psi(\mathbf{X}_{(k-1)\Delta T}, \Delta T, Z_k);$ 
    
```

For example, for a stochastic differential equation of the form

$$d\mathbf{X}_t = b(\mathbf{X}_t)dt + \sigma(\mathbf{X}_t)dW_t, \quad (28)$$

the Euler scheme corresponds to the choice

$$\psi(\mathbf{x}, t, z) = \mathbf{x} + b(\mathbf{x})t + \sigma(\mathbf{x})\sqrt{t}z. \quad (29)$$

In this section, we choose continuous functions $b, \sigma : \mathbb{R} \longrightarrow \mathbb{R}$ with at most linear growth, such that $\sigma(0) = 0$ and $b(0) \geq 0$, and simulate a trajectory solving the N -dimensional equation (3). We use a long time horizon, for example $T = 100$, because we wish to verify numerically that the solution remains in the domain $\mathcal{D}$ at all times. In the next section, the simulations use the Euler scheme associated with (3). We denote the number of time steps by n and the number of simulations by m .

Remark 3.1. We know that, if $Y_0 \geq 0$, the Volterra process $Y_t = \boldsymbol{\gamma}^\top \mathbf{Y}_t$ remains nonnegative over time. Therefore, when b and/or σ are defined only on $\mathbb{R}_+$, the Euler scheme will use the positive part of the Volterra process.

3.2 Simulations

We perform simulations for $N = 2$ and $N = 3$, choosing respectively $\boldsymbol{\gamma}^\top = (1, 2)$, $\boldsymbol{\rho}^\top = (1, 5)$ and $\boldsymbol{\gamma}^\top = (1, 2, 3)$, $\boldsymbol{\rho}^\top = (1, 5, 25)$.

3.2.1 Square-Root Model

Take $b(x) = \theta - \lambda x$ and $\sigma(x) = \nu\sqrt{x}$. This yields the multifactor square-root model

$$d\mathbf{V}_t^N = -\text{diag}(\boldsymbol{\rho})(\mathbf{V}_t^N - \mathbf{v}_0)dt + (\theta - \lambda\boldsymbol{\gamma}^\top \mathbf{V}_t^N)\mathbf{1}dt + \nu\sqrt{\boldsymbol{\gamma}^\top \mathbf{V}_t^N}\mathbf{1}dW_t.$$

The functions b and σ satisfy the assumptions of Theorem 2.1. Indeed, to show that σ has at most linear growth, it is enough to write

$$\forall z \geq 0, \quad \sigma(z) = \nu\sqrt{z} \leq \nu(1 + z)$$

Figure 1 shows trajectories in this model.

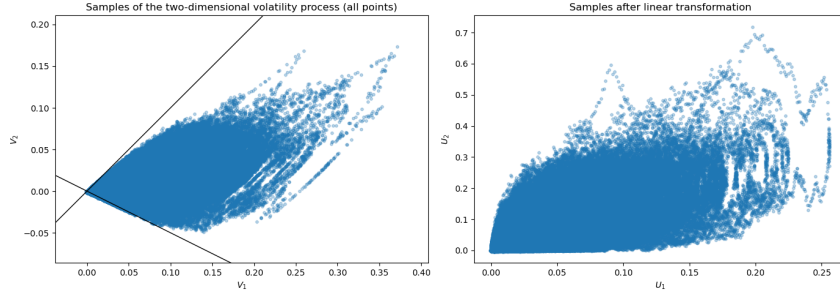


Figure 1: Trajectories of the square-root model (the original process on the left and the same process left-multiplied by Q on the right). The chosen parameters are $\lambda = 0.3$, $\nu = 0.3$, $\theta = 0.02$, $\mathbf{V}_0 = 0$, $\mathbf{v}_0 = 0$, $m = 30$, $n = 10^5$ and $T = 100$.

Figure 2 shows the same plot with only the terminal points.

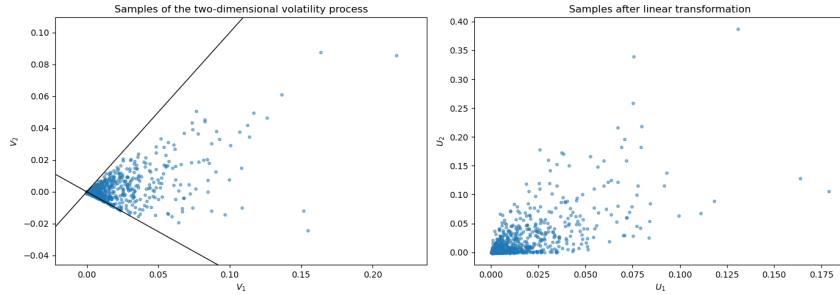
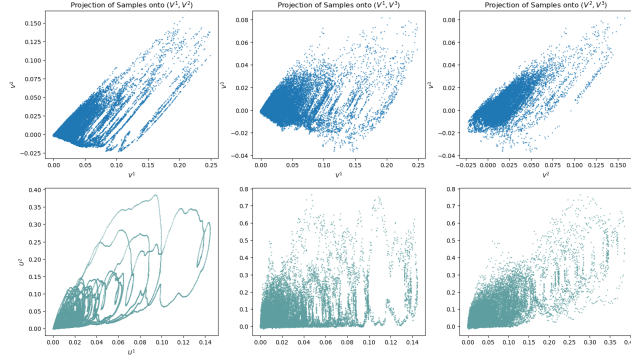


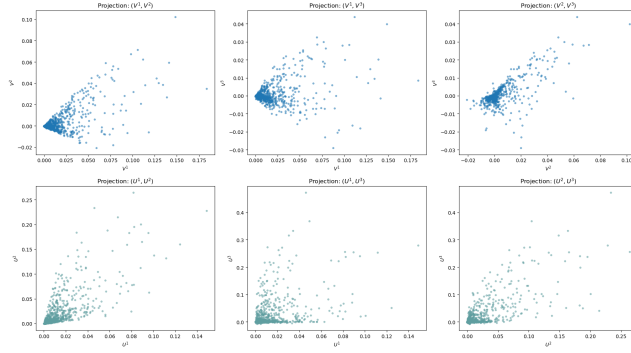
Figure 2: Terminal points of trajectories of the square-root model. We chose $m = 10^3$; the other parameters are the same as in Figure 1.

Figures 3a and 3b show the same quantities for $N = 3$. For easier visualisation, we show the three projections onto pairs of coordinates (the original process on top and the process left-multiplied by Q below).

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(a) Trajectories



(b) Terminal points

Figure 3: Trajectories and terminal points of the square-root model. The parameters are those used in Figure 1.

3.2.2 Model with a Power Diffusion Function σ

A more general version of the preceding square-root model is obtained when σ has the form

$$\sigma(z) = \nu z^\eta$$

for $\eta \in [\frac{1}{2}, 1]$. We run simulations for several values of η , namely $\eta \in \{0.6, 0.75, 0.9, 0.95\}$. The left column of Figure 4 shows 30 simulated trajectories for each value of η (the original process on the left and the process left-multiplied by Q on the right). The right column of Figure 4 shows 1000 simulated terminal points for each value of η (again, the original process on the left and the process left-multiplied by Q on the right).

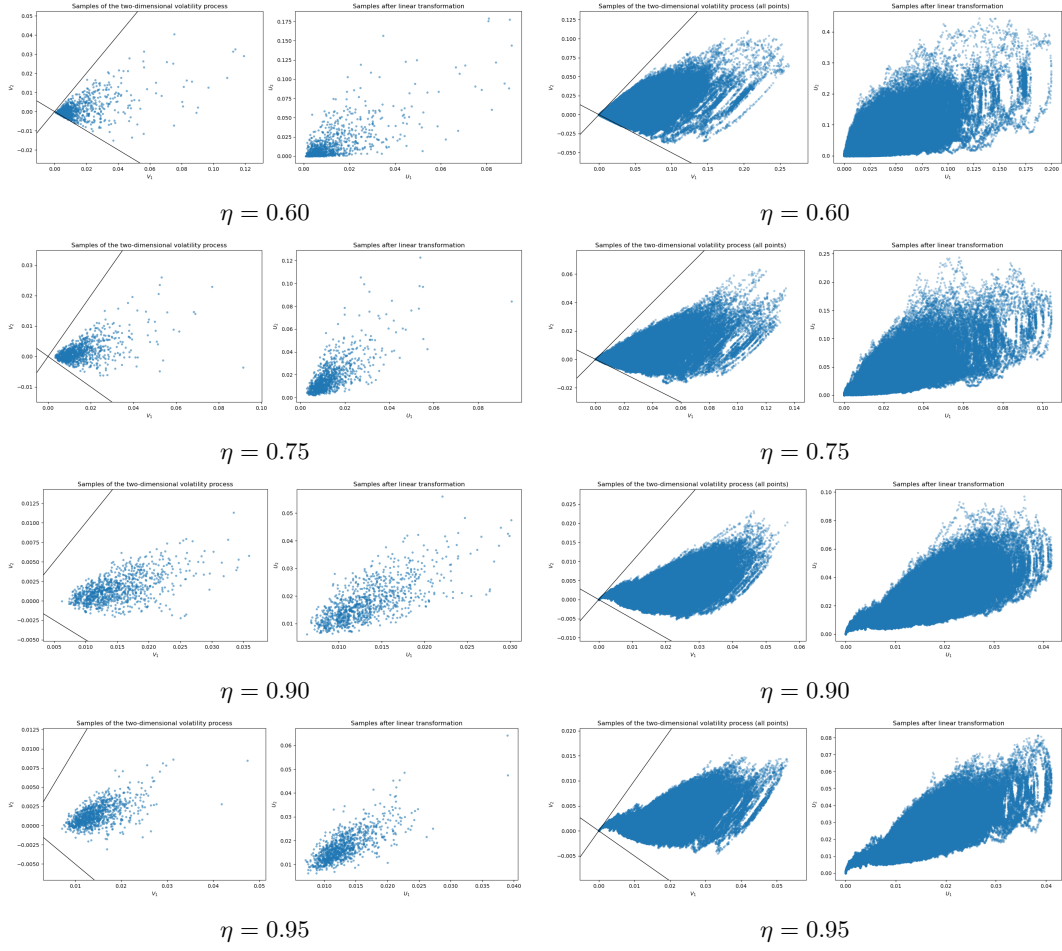


Figure 4: Trajectories (left) and terminal points (right) of the model for different values of η . The chosen parameters are $\lambda = 0.3$, $\nu = 0.3$, $\theta = 0.02$ and $n = 10^5$. We used $m = 30$ for the trajectories and $m = 10^3$ for the terminal points.

We observe that the closer η is to 1, the less the trajectories cling to the boundary of the domain $\mathcal{D}$. Moreover, the trajectories tend to move away from the boundary of $\mathcal{D}$ over time, increasingly so as η approaches 1.

3.2.3 Model with Hyperbolic-Tangent μ and Sigmoid σ

The chosen model is given by

$$b(z) = \theta(\nu - z) \tanh(\beta z), \quad \sigma(z) = \frac{\alpha z}{1 + \gamma|z|}.$$

Figure 5 shows trajectories in this model.

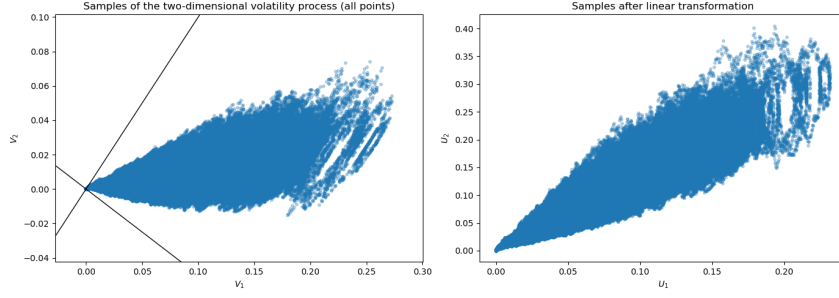


Figure 5: Model trajectories (the original process on the left and the process left-multiplied by Q on the right). The chosen parameters are $\theta = 0.5, \beta = 1.5, \alpha = 0.3, \gamma = 1, n = 10^5, T = 100$ and $m = 30$.

3.2.4 Black–Scholes Model

Another model to consider is the Black–Scholes model, defined by

$$b(z) = \mu z, \quad \sigma(z) = \sigma z.$$

Figure 6 shows trajectories in this model.

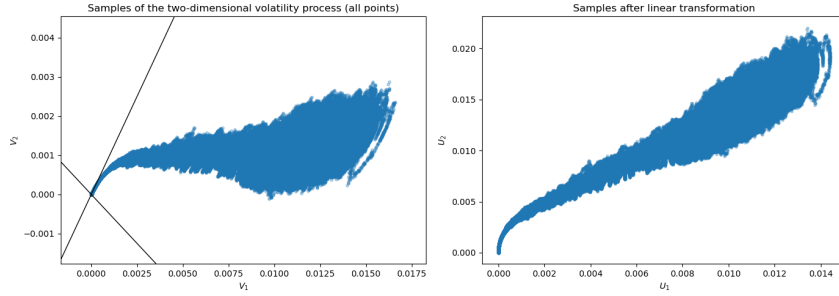


Figure 6: Model trajectories (the original process on the left and the process left-multiplied by Q on the right). The chosen parameters are $\mu = 0.08, \nu = 0.1, n = 10^5, T = 100$ and $m = 30$.

3.3 Construction of a Simulation Scheme for the Square-Root Model via Splitting

3.3.1 Splitting

So far, we have used the Euler scheme for the simulations. In this part, we study another scheme and compare it with the Euler scheme for the square-root model.

We again start from a solution X of the Volterra equation

$$X_t = x_0 + \int_0^t K(t-s)b(X_s)ds + \int_0^t K(t-s)\sigma(X_s)dW_s \quad (30)$$

where $K(t) = \sum_{i=1}^N \gamma_i e^{-\rho_i t}$. We can write X as $X_t = x_0 + \sum_{i=1}^N \gamma_i X_t^i$, where the X^i satisfy

$$dX_t^i = -\rho_i X_t^i dt + (\theta - \lambda X_t)dt + \nu \sqrt{X_t} dW_t, \quad \rho_0^i = 0 \quad \forall i \in \llbracket 1, N \rrbracket. \quad (31)$$

We use a splitting technique to construct a simulation scheme for the N -dimensional equa-

tion (31). Indeed, the infinitesimal generator of equation (31) is

$$\mathcal{L}f(x) = \sum_{i=1}^N (\theta - \lambda x - \rho_i) \partial_i f(x) + \frac{1}{2} \sum_{i,j \in \llbracket 1, N \rrbracket} \nu^2 X_t \partial_{ij} f(x).$$

We write it as

$$\mathcal{L} = \mathcal{L}_1 + \mathcal{L}_2$$

where $\mathcal{L}_1 f = -\sum_{i=1}^N \rho_i \partial_i f$ is associated with

$$dX_t^i = -\rho_i X_t^i dt \quad (32)$$

and $\mathcal{L}_2$ is the infinitesimal generator associated with

$$dX_t^i = (\theta - \lambda X_t) dt + \nu \sqrt{X_t} dW_t \quad \text{with } X_t = x_0 + \sum_{i=1}^N \gamma_i X_t^i. \quad (33)$$

We construct a simulation scheme for each of the two equations and then derive a scheme for equation (31). This is straightforward for equation (32), since the deterministic equation can be solved exactly. We use the exact solution as the scheme:

$$\forall \mathbf{x} = (x^1, \dots, x^N) \in \mathbb{R}^N, \quad \psi_1(\mathbf{x}, t) = (x^1 e^{-\rho_1 t}, \dots, x^N e^{-\rho_N t}).$$

For equation (33), introduce ξ_t^x , a solution of the following constant-kernel Volterra equation:

$$\xi_t^x = x + \int_0^t K(0)(\theta - \lambda \xi_s^x) ds + \int_0^t K(0) \nu \sqrt{\xi_s^x} dW_s. \quad (34)$$

It can be rewritten as

$$\xi_t^x = x + \int_0^t (\bar{\theta} - \bar{\lambda} \xi_s^x) ds + \int_0^t \bar{\nu} \sqrt{\xi_s^x} dW_s \quad (35)$$

where $\bar{\theta} = K(0)\theta$, $\bar{\lambda} = K(0)\lambda$ and $\bar{\nu} = K(0)\nu$.

This is a Cox–Ingersoll–Ross (CIR) process. One can show that the solution of (33) is given by $A_{\mathbf{x}}(\xi_t^x)$, with

$$A_{\mathbf{x}}(y) = \left(x_1 + \frac{y - x}{K(0)}, \dots, x_N + \frac{y - x}{K(0)} \right)$$

where $x = x_0 + \sum_{i=1}^N \gamma_i x^i$.

Thus, equation (33) is only apparently N -dimensional: solving it merely requires solving the one-dimensional equation (34). The processes X^i solving (33) differ only by constants. This is verified numerically in Figure 7.

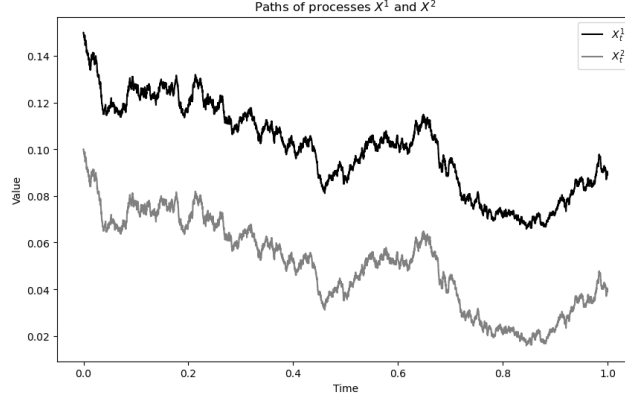


Figure 7: Multifactor process as a function of time in dimension $N = 2$. The parameters are $a = 0.02$, $\kappa = 0.3$, $\nu = 0.3$, $\rho = (1, 10)$, $\gamma = (1, 2)$, $\mathbf{X}_0 = (0.05, 0.1)$, $T = 1$ and $n = 10^4$.

3.3.2 Simulation Scheme for the CIR Process

We construct a simulation scheme for the CIR process in order to derive a simulation scheme for equation (33). Several schemes are presented in this subsection.

The CIR process is well known. Its transition law is a noncentral chi-squared distribution. Indeed, one can show that

$$\xi_{t+\Delta t}^x \mid \xi_t^x \sim c \chi_d^2(\lambda)$$

$$\text{with } d = \frac{4\bar{\theta}}{\bar{\lambda}\bar{\nu}^2}, \quad \lambda = \frac{4\bar{\lambda}\xi_t^x e^{-\bar{\lambda}\Delta t}}{\bar{\nu}^2(1 - e^{-\bar{\lambda}\Delta t})} \quad \text{and} \quad c = \frac{\bar{\nu}^2(1 - e^{-\bar{\lambda}\Delta t})}{4\bar{\lambda}}.$$

This provides a first method for simulating ξ_t^x .

Another approach is to use the scheme $\varphi(x, t, U)$ from [2], defined by

$$\varphi(x, t, U) = \begin{cases} e^{-\frac{\bar{\lambda}t}{2}} \left(\sqrt{\bar{\theta} - \frac{\bar{\nu}^2}{4}} \zeta_{\bar{\lambda}} \left(\frac{t}{2} \right) + e^{-\frac{\bar{\lambda}t}{2}} x + \frac{\bar{\nu}}{2} \sqrt{t} w(U) \right)^2 & \text{if } x \geq \bar{K}_2(t) \\ + \left(\bar{\theta} - \frac{\bar{\nu}^2}{4} \right) \zeta_{\bar{\lambda}} \left(\frac{t}{2} \right) & \\ \mathbf{1}_{\{U \leq \pi(t, x)\}} \cdot \frac{\bar{u}_1(t, x)}{2\pi(t, x)} + \mathbf{1}_{\{U > \pi(t, x)\}} \cdot \frac{\bar{u}_1(t, x)}{2(1 - \pi(t, x))} & \text{if } x < \bar{K}_2(t) \end{cases}$$

and

$$\zeta_k(t) = \frac{1 - e^{-kt}}{k} \quad \text{for } k \neq 0 \quad \text{and} \quad \zeta_0(t) = t,$$

$$w(u) = \sqrt{3} (\mathbf{1}_{\{u > 5/6\}} - \mathbf{1}_{\{u \leq 1/6\}}), \quad \pi(t, x) = \frac{1 - \sqrt{1 - \frac{\bar{u}_1(t, x)^2}{\bar{u}_2(t, x)}}}{2},$$

with

$$\bar{u}_1(t, x) = x e^{-\bar{\lambda}t} + \bar{\theta} \zeta_{\bar{\lambda}}(t), \quad \bar{u}_2(t, x) = \bar{u}_1(t, x)^2 + \bar{\nu}^2 \zeta_{\bar{\lambda}}(t) \left[\bar{\theta} \zeta_{\bar{\lambda}}(t)/2 + x e^{-\bar{\lambda}t} \right],$$

and

$$\begin{aligned} \bar{K}_2(t) = & \mathbf{1}_{\{\bar{\nu}^2 > 4\bar{\theta}\}} e^{\frac{\bar{\lambda}t}{2}} \left(\left(\frac{\bar{\nu}^2}{4} - \bar{\theta} \right) \zeta_{\bar{\lambda}} \left(\frac{t}{2} \right) \right. \\ & \left. + \left[\sqrt{e^{\frac{\bar{\lambda}t}{2}} \left(\frac{\bar{\nu}^2}{4} - \bar{\theta} \right) \zeta_{\bar{\lambda}} \left(\frac{t}{2} \right)} + \frac{\bar{\nu}}{2} \sqrt{3t} \right]^2 \right). \end{aligned}$$

According to [2], the scheme φ is second-order.

At this stage, we have three schemes for the process ξ_t^x : the Euler scheme, the scheme based on the transition law of the CIR process, and the scheme φ from [2]. Figure 8 shows three simulations of a CIR process using the scheme φ . Figure 9 gives the empirical distribution of ξ_T^x defined by (34), together with the computation times for each of the three simulation methods.

Remark 3.2. *In Figure 9, two approaches were used to estimate the distribution of ξ_T^x empirically using the transition law of ξ_t^x . The first consists in simulating ξ_T^x directly in a single step from its exact distribution. The second approach uses a discretisation of the time interval $[0, T]$, in which case N successive transitions of the process are simulated.*

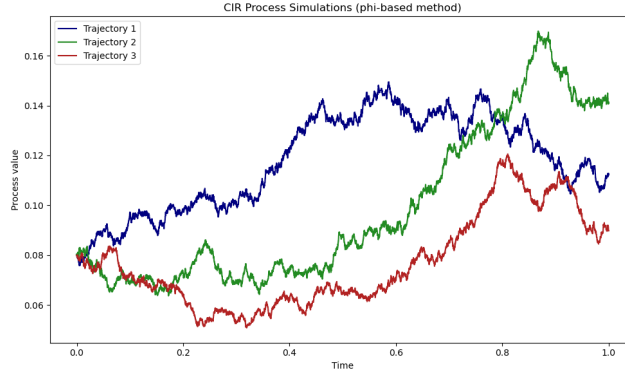


Figure 8: Three simulations of a CIR process. The chosen parameters are $\theta = 0.05$, $\lambda = 0.15$, $\nu = 0.1$, $x = 0.08$, $\gamma = (0.96, 0.07, 0.04, 0.03, 0.02)$, $T = 1$ and $n = 10^4$.

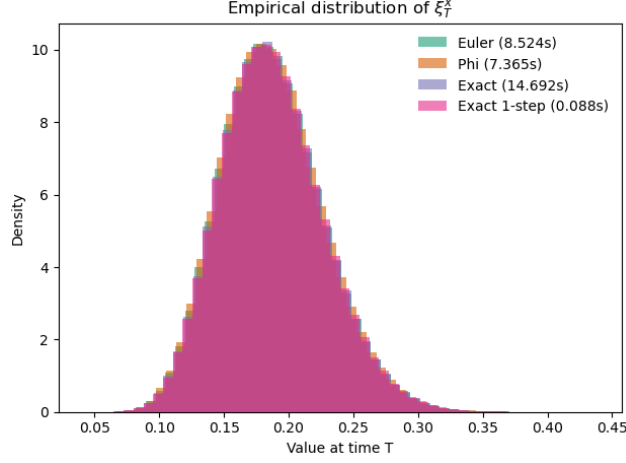


Figure 9: Empirical distribution of ξ_T^x . The chosen parameters are $\theta = 0.1$, $\lambda = 0.15$, $\nu = 0.1$, $x = 0.1$, $T = 1$, $n = 10^2$ and $m = 10^6$. The vector γ is the same as in Figure 8.

For each of the three methods, Figure 10 shows the time required to generate $m = 10^3$ trajectories as a function of the time step $\Delta t = T/N$. The Euler method is the fastest, followed by the scheme φ , while the exact method based on the transition law of the CIR process is the slowest.

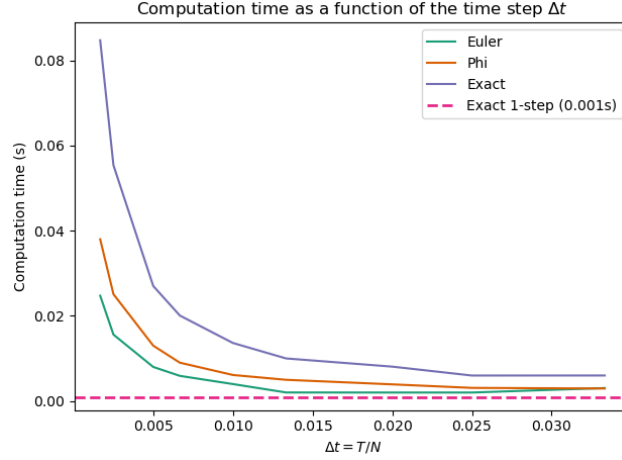


Figure 10: Comparison of simulation times. The chosen parameters are those used in Figure 9.

3.4 Zero-Coupon Bond in the CIR Model

In this section, we present an application of CIR process simulation to zero-coupon bond pricing. We begin by recalling the definition of a zero-coupon bond.

Definition 3.1 (Zero-coupon bond). *A zero-coupon bond is a security entitling its holder to one euro at a maturity date T . We denote the value of this security at time $t \in [0, T]$ by $P(t, T)$.*

Remark 3.3. *In particular, $P(T, T) = 1$, meaning that at maturity the value of the bond is exactly equal to the amount it pays.*

Let $\mathbb{P}^*$ be a probability measure under which $\tilde{P}(t, T) = \left(e^{-\int_0^t r(s) ds} P(t, T) \right)_{t \in [0, T]}$ is a martingale. The process $(r(t))_{t \in [0, T]}$ is called the instantaneous interest rate. It is $(\mathcal{F}_t)_{t \geq 0}$ -adapted, and its dynamics are given by

$$dr(t) = (a - br(t))dt + \sigma\sqrt{r(t)}dW_t \quad (36)$$

where $(W_t)_{t \geq 0}$ is a Brownian motion under $\mathbb{P}^*$. The process $(r(t))_{t \geq 0}$ is a Vasicek process.

Recall that, according to [8], the zero-coupon bond price is then given by

$$P(t, T) = \mathbb{E}^* \left(e^{-\int_t^T r(s) ds} \mid \mathcal{F}_t \right). \quad (37)$$

We use the three simulation schemes for $r(t)$ defined in the preceding sections to compute the initial zero-coupon bond price $P(0, T)$ by Monte Carlo. To this end, we simulate a large number m of CIR process trajectories, denoted by $r^1, \dots, r^m$. We use the trapezoidal rule to approximate the integral

$$\int_0^T r^i(s) ds \simeq \frac{T}{N} \left(\frac{r^i(0) + r^i(T)}{2} + \sum_{l=1}^{n-1} r^i(t_l) \right) \quad (38)$$

where $t_k = kT/N$. The trapezoidal rule is second-order, which allows us to retain the second-order accuracy of the scheme φ . We therefore approximate the zero-coupon bond price by

$$P(0, T) \simeq \frac{1}{m} \sum_{i=1}^m \exp \left(-\frac{T}{N} \left(\frac{r^i(0) + r^i(T)}{2} + \sum_{l=1}^{n-1} r^i(t_l) \right) \right). \quad (39)$$

According to [8], the zero-coupon bond price actually has the following closed-form expression:

$$P(t, T) = \exp(-a\phi(T-t) - r(t)\psi(T-t)) \quad (40)$$

with

$$\phi(t) - \frac{2}{\sigma^2} \log \left(\frac{2\gamma e^{\frac{t(\gamma+b)}{2}}}{\gamma - b + e^{\gamma t}(\gamma + b)} \right) \quad \text{and} \quad \psi(t) = \frac{2(e^{\gamma t} - 1)}{\gamma - b + e^{\gamma t}(\gamma + b)}. \quad (41)$$

In particular,

$$P(0, T) = \exp(-a\phi(T) - r_0\psi(T)).$$

Table 1 gives the zero-coupon bond price obtained by Monte Carlo for the three schemes, together with the exact price obtained from the closed-form expression.

Method	$P(0, T)$	95% CI	CI width	Absolute error	Relative error (%)	Time (s)
Exact method	0.621587	[0.621533, 0.621641]	0.000108	0.001019	0.164216	4.12
Scheme φ	0.621476	[0.621423, 0.621529]	0.000106	0.000908	0.146377	2.51
Euler scheme	0.613459	[0.613400, 0.613519]	0.000120	0.007108	1.145449	3.76
Closed-form expression	0.620568	—	—	—	—	Instantaneous

Table 1: Comparison of zero-coupon bond pricing methods. The parameters are $a = 0.5$, $k = 0.7$, $\sigma = 0.5$, $r_0 = 0.4$, $T = 1$, $N = 2$ and $m = 10^7$.

Figure 11 shows the zero-coupon bond price obtained by Monte Carlo using simulations from the scheme φ and from the exact transition law of the CIR process. The graph shows quadratic convergence of this price to the price calculated from the closed-form expression.

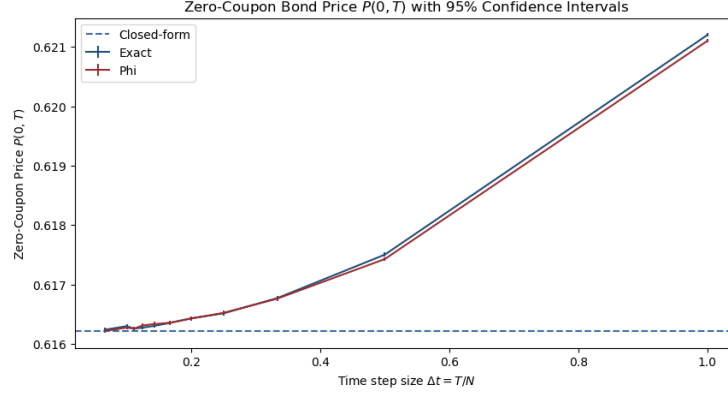


Figure 11: Convergence of the price calculated by Monte Carlo using the scheme φ and the exact transition law of the CIR process. We choose $a = 0.5$, $k = 0.7$, $\sigma = 0.5$, $\xi_0^x = 0.4$, $T = 1$ and $m = 10^7$.

3.4.1 Simulation Scheme for the Square-Root Model

For $\mathbf{x} = (x^1, \dots, x^N)$, set $\psi_2(\mathbf{x}, t, U) = A_{\mathbf{x}}(\varphi(\mathbf{x}, t, U))$, where $x = x_0 + \sum_{i=1}^N \gamma_i x^i$. We also denote by

$$C_{\text{pol}}^\infty = \{f : \mathbb{R} \rightarrow \mathbb{R} \mid \forall \alpha \in \mathbb{N}, \exists C_\alpha > 0, e_\alpha \in \mathbb{N}^*, \forall x \in \mathbb{R}, |\partial_\alpha f(x)| \leq C_\alpha(1 + |x|^{e_\alpha})\}$$

the set of C^∞ functions whose derivatives have at most polynomial growth.

The following theorem, from [3], provides a simulation scheme for the square-root model.

Theorem 3.1. *Define*

$$\psi(\mathbf{x}, t, U) = \psi_1(\psi_2(\psi_1(\mathbf{x}, t/2), t, U), t/2).$$

Consider the scheme defined by $\hat{\mathbf{X}}_0 = 0$ and $\hat{\mathbf{X}}_{l\frac{T}{N}} = \psi\left(\hat{\mathbf{X}}_{(l-1)\frac{T}{N}}, \frac{T}{N}, U_l\right)$ for $l \geq 1$, where $(U_l)_{l \geq 1}$ are i.i.d. uniform random variables on $[0, 1]$. Set $\hat{X}_{l\frac{T}{N}} = x_0 + \sum_{i=1}^N \gamma_i \hat{\mathbf{X}}_{l\frac{T}{N}}^i$. Then

$$\forall f \in C_{\text{pol}}^\infty(\mathbb{R}), \exists K > 0, \exists N_0 \in \mathbb{N}^*, \forall N \geq N_0, |\mathbb{E}(f(\hat{X}_T)) - \mathbb{E}(f(X_T))| \leq \frac{K}{N^2}.$$

Figure 12 shows a trajectory obtained using the preceding scheme and one obtained using the Euler method. The trajectory simulated with the scheme ψ follows the boundary of the domain $\mathcal{D}$ more closely. This indicates that the scheme ψ preserves the domain better than the Euler scheme.

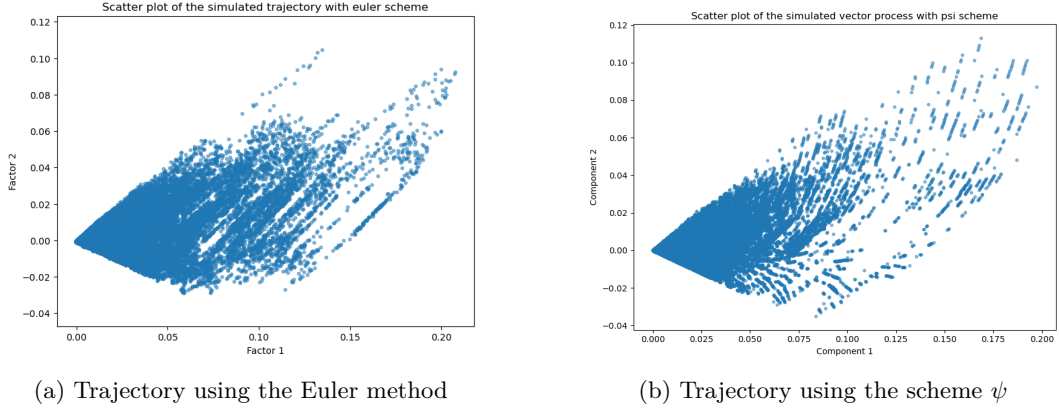


Figure 12: Simulation of the multifactor process using the two schemes. The chosen parameters are $a = 0.02$, $k = 0.3$, $\sigma = 0.3$, $\rho_0 = 0$, $T = 10^2$, $n = 10^5$, $\boldsymbol{\rho} = (1, 10)$ and $\boldsymbol{\gamma} = (1, 2)$.

Figure 13 shows the terminal points of a large number of trajectories for the two simulation schemes.

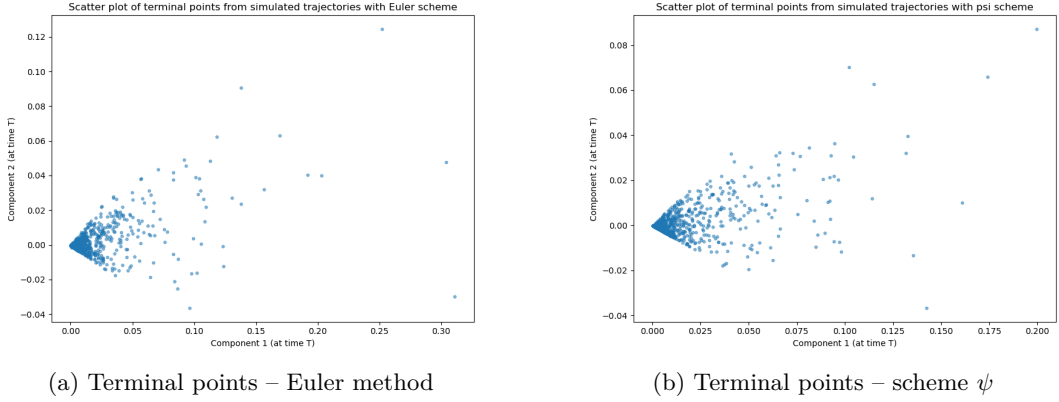


Figure 13: Terminal points of trajectories. The chosen parameters are $n = 10^4$ and $m = 10^4$. The other parameters are the same as in Figure 12.

Figure 14 shows the time required to simulate a multifactor trajectory as a function of the time step. The Euler scheme is faster.

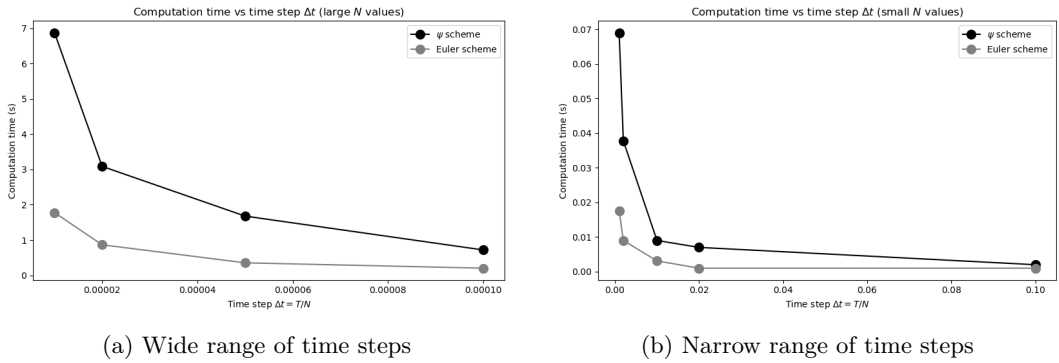


Figure 14: Simulation times for the multifactor processes. The chosen time horizon is $T = 1$. The other parameters are those used in Figure 13.

3.5 Scheme for the Heston Model

We repeat the preceding analysis, this time for the Heston model. The Heston model is defined by

$$\begin{cases} dY_t = (r - \frac{1}{2}X_t) dt + \sqrt{X_t} (\rho dW_t + \sqrt{1 - \rho^2} dW_t^\perp) \\ \text{with } X_t = x_0 + \sum_{i=1}^N \gamma_i X_t^i, \\ dX_t^i = -\rho_i X_t^i dt + (a - kX_t) dt + \sigma \sqrt{X_t} dW_t, \quad \forall i \in \llbracket 1, N \rrbracket. \end{cases} \quad (42)$$

We use a splitting method to simulate X_t^i . Let $\mathcal{L}$ denote the infinitesimal generator of the preceding equation in dimension $N+1$. We write $\mathcal{L} = \mathcal{L}_1 + \mathcal{L}_2$, where $\mathcal{L}_1$ is the infinitesimal generator associated with the ODE

$$\begin{cases} dX_t^i = -\rho_i X_t^i dt, \\ dY_t = 0. \end{cases} \quad (43)$$

and $\mathcal{L}_2$ is associated with

$$\begin{cases} dX_t^i = (a - kX_t) dt + \sigma \sqrt{X_t} dW_t, \quad \forall i \in [1, n] \\ \text{with } X_t = x_0 + \sum_{i=1}^n \gamma_i X_t^i, \\ dY_t = (r - \frac{1}{2}X_t) dt + \sqrt{X_t} (\rho dW_t + \sqrt{1 - \rho^2} dW_t^\top). \end{cases} \quad (44)$$

The ODE associated with $\mathcal{L}_1$ can be solved exactly, and its solution is given by the scheme

$$\bar{\psi}_1((\mathbf{x}, y), t) = (\psi_1(\mathbf{x}, t), y)$$

For the equation associated with $\mathcal{L}_2$, we use the scheme $\bar{\psi}_2$ given in [2].

The scheme used for the multifactor Heston model is therefore

$$\bar{\Psi}((\mathbf{x}, y), t, Z) = \bar{\psi}_1(\bar{\psi}_2(\bar{\psi}_1((\mathbf{x}, y), t/2), t, Z), t/2).$$

Figure 15 shows a trajectory of the multifactor Heston process in dimension 2. Figure 16 shows terminal points.

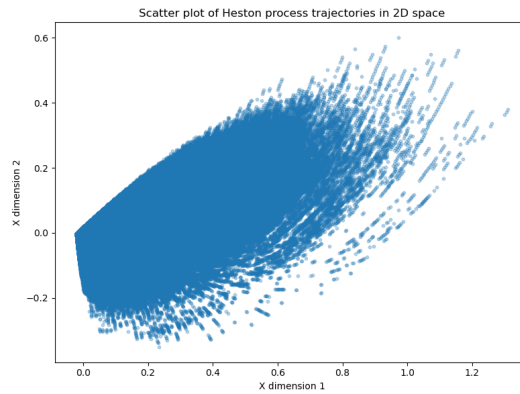


Figure 15: A trajectory of the multifactor Heston process. The chosen parameters are $S_0 = 1$, $r = 0$, $\rho = -0.7$, $\rho_0 = 0.02$, $a = 0.02$, $k = 0.3$, $\sigma = 0.3$, $\boldsymbol{\rho} = (0.06, 1.1)$, $\boldsymbol{\gamma} = (0.9, 0.08)$, $T = 100$ and $n = 10^4$.

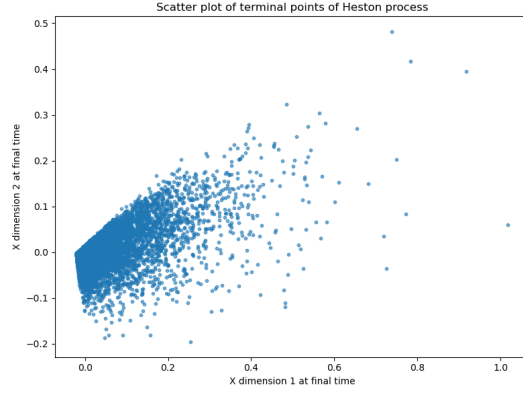


Figure 16: Terminal points of trajectories of the multifactor Heston process. We choose $m = 10^4$; the other parameters are as in Figure 15.

In this model, the risky asset price is $S_t = e^{Y_t}$. Figure 17 shows several simulated paths of the risky asset price. The simulations use the scheme described above.

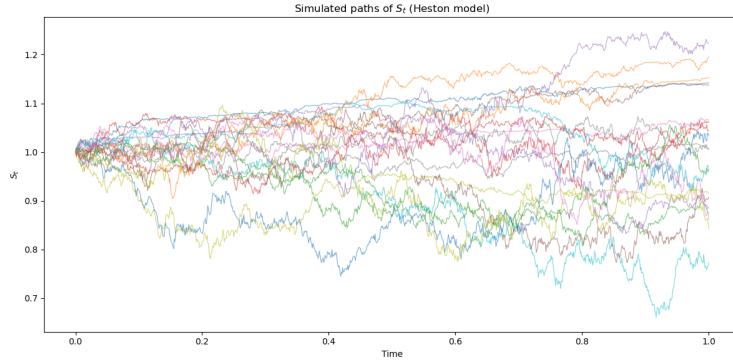


Figure 17: Several simulated paths of the risky asset price. The chosen parameters are $\boldsymbol{\rho} = (0.06, 1.04, 1.78, 2.52, 3.24)$, $\boldsymbol{\gamma} = (0.95, 0.06, 0.04, 0.03, 0.02)$, $S_0 = 1$, $\rho = -0.7$, $x_0 = 0.02$, $a = 0.02$, $k = 0.3$, $\sigma = 0.3$, $T = 1$ and $n = 10^3$.

Figure 18 shows the empirical values of $\mathbb{E}(e^{-X_T/\rho_0})$, $\mathbb{E}((e^{Y_T} - S_0)_+)$ and $\mathbb{E}\left(\frac{X_T}{\rho_0} (e^{-Y_T} - S_0)_+\right)$ as functions of T/N , with a 95% confidence interval at each point.

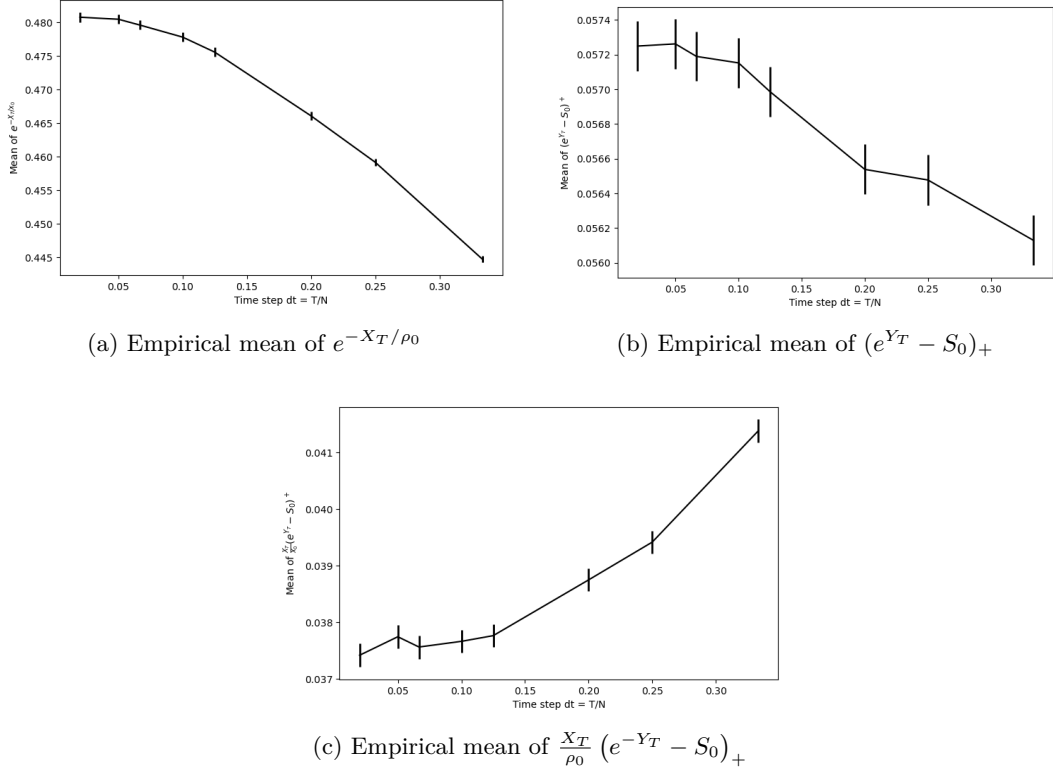


Figure 18: Convergence of the three empirical means as functions of the time step T/N . The chosen parameters are those used in Figure 18b.

3.6 Jacobi Version

In this section, we consider another local-volatility model: the Jacobi model. It is similar to the Heston model, except that $\sqrt{Q(V_t)}$ is used instead of $\sqrt{V_t}$, where Q is a quadratic function.

3.6.1 Presentation of the Jacobi Model

We present the Jacobi model in this section. Let $0 \leq v_{\min} < v_{\max}$ and define

$$\forall v \in [v_{\min}, v_{\max}], \quad Q(v) = \frac{(v - v_{\min})(v_{\max} - v)}{(\sqrt{v_{\max}} - \sqrt{v_{\min}})^2}. \quad (45)$$

Remark 3.4. The function Q satisfies $Q(v) \in [0, v]$ for every $v \in [v_{\min}, v_{\max}]$, because

$$v - Q(v) = \frac{(v - \sqrt{v_{\min}v_{\max}})^2}{(\sqrt{v_{\max}} - \sqrt{v_{\min}})^2} \geq 0$$

Consider the process (V_t, X_t) satisfying

$$\begin{cases} dV_t = \kappa(\theta - V_t)dt + \sigma\sqrt{Q(V_t)}dW_{1t} \\ dX_t = (r - \delta - \frac{V_t}{2})dt + \rho\sqrt{Q(V_t)}dW_{1t} + \sqrt{V_t - \rho^2Q(V_t)}dW_{2t}. \end{cases} \quad (46)$$

where $\kappa > 0$, $\theta \in [v_{\min}, v_{\max}]$ is the long-run variance, $\sigma > 0$ is the volatility of variance, r is the interest rate, $\rho \in [-1, 1]$, and $(W_{1t})_{t \in [0, T]}$ and $(W_{2t})_{t \in [0, T]}$ are two independent Brownian motions defined on a probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \in [0, T]}, \mathbb{P})$. System (46) defines

the Jacobi model. According to [1], the process (V_t, X_t) is well defined: for every $(V_0, X_0) \in [v_{\min}, v_{\max}] \times \mathbb{R}$, there exists a unique process (V_t, X_t) solving (46), starting from (V_0, X_0) and taking values in $[v_{\min}, v_{\max}] \times \mathbb{R}$. Further details on this model can be found in [1].

Remark 3.5. *The Jacobi model generalises the Heston model in the sense that the latter is recovered in the limit. Indeed, if $v_{\min} = 0$, then $Q(v) \rightarrow v$ as $v_{\max} \rightarrow +\infty$.*

In the Jacobi model, the risky asset price is $S_t = e^{X_t}$. The process $e^{-(r-\delta)t}S_t$ is a martingale. Indeed, Itô's formula gives

$$d\left(e^{-(r-\delta)t}S_t\right) = e^{-(r-\delta)t+X_t} \left(\rho\sqrt{Q(V_t)}dW_{1t} + \sqrt{V_t - \rho^2Q(V_t)}dW_{2t}\right)$$

3.6.2 Pricing European Options

Consider a European option with payoff $f(X_T)$ at maturity T . Define

$$w(x) := \frac{1}{\sqrt{2\pi\sigma_w^2}} e^{\frac{-(x-\mu_w)^2}{2\sigma_w^2}}.$$

Define the Hermite polynomials (probabilists' convention) by

$$\mathcal{H}_n(x) = (-1)^n e^{x^2/2} \frac{d^n}{dx^n} e^{-x^2/2}. \quad (47)$$

We also define the Hilbert space

$$L_w^2 = \left\{ f \mid \|f\|_w^2 = \int_{\mathbb{R}} f(x)^2 w(x) dx < +\infty \right\} \quad (48)$$

equipped with the inner product $(f, g)_w = \int_{\mathbb{R}} f(x)g(x)w(x)dx$. The sequence $(H_n)_{n \geq 0}$ is a Hilbert basis of L_w^2 , where

$$H_n(x) = \frac{1}{\sqrt{n!}} \mathcal{H}_n\left(\frac{x - \mu_w}{\sigma_w}\right). \quad (49)$$

Set $\ell = g_T/w$. According to [1], if

$$\mathbb{E} \left(\left(\int_0^T (V_s - \rho^2 Q(V_s)) ds \right)^{-1/2} \right) < +\infty \quad (50)$$

then $\ell \in L_w^2$. We therefore assume that Condition (50) holds and that $f \in L_w^2$, so that the price π_f of the option with payoff $f(X_T)$ at maturity T is well defined and given by

$$\pi_f = \int_{\mathbb{R}} f(x)g_T(x)dx = (f, \ell)_w = \sum_{n \geq 0} f_n \ell_n \quad (51)$$

where $f_n = (f, H_n)_w$. We approximate the price π_f by truncating its series at order N , that is, by considering

$$\pi_f^N = \sum_{n=0}^N f_n \ell_n. \quad (52)$$

The approximate price π_f^N then converges to the true price π_f as N tends to infinity. Pricing therefore requires the coefficients f_n and ℓ_n to be computed for every $n \geq 0$.

Remark 3.6. *Only the Fourier coefficients f_n depend on the option. The Hermite moments apply to all European options.*

3.6.3 Calculation of the Fourier Coefficients

For the European call $f_{\text{Call}}(x) = e^{-rT} (e^x - e^k)_+$ and the digital option $f_{\text{Digit}}(x) = e^{-rT} \mathbf{1}_{[k, +\infty[}(x)$, the Fourier coefficients were calculated in [1], to which we refer the reader.

3.6.4 Calculation of the Hermite Moments

In this section, we show how to calculate the Hermite moments ℓ_n . Let $\text{Pol}(\mathbb{R}^d)$ denote the space of polynomials on $\mathbb{R}^d$, and let $\deg(p)$ denote the degree of a polynomial $p \in \text{Pol}(\mathbb{R}^d)$. Let $\text{Pol}_N(\mathbb{R}^d)$ be the set of polynomials $p \in \text{Pol}(\mathbb{R}^d)$ such that $\deg(p) \leq N$, and set $M = \dim \text{Pol}_N(\mathbb{R}^d)$. A counting argument gives $M = \binom{N+d}{d}$.

Let $\mathcal{G}$ denote the infinitesimal generator of (46). We have

$$\mathcal{G}f(x, v) = b(v)^\top \nabla f(v, x) + \frac{1}{2} \text{Tr} (a(v) \nabla^2 f(v, x)) \quad (53)$$

with

$$b(v) = \begin{pmatrix} \kappa(\theta - v) \\ r - \delta - v/2 \end{pmatrix} \quad \text{and} \quad a(v) = \begin{pmatrix} \sigma^2 Q(v) & \rho \sigma Q(v) \\ \rho \sigma Q(v) & v \end{pmatrix}. \quad (54)$$

Remark 3.7. The infinitesimal generator $\mathcal{G}$ is linear and preserves $\text{Pol}_N(\mathbb{R}^2)$. Therefore, the map

$$\mathcal{G} : \text{Pol}_N(\mathbb{R}^2) \longrightarrow \text{Pol}_N(\mathbb{R}^2)$$

is an endomorphism of the vector space $\text{Pol}_N(\mathbb{R}^2)$.

Let

$$\pi : \mathcal{E}_N \longrightarrow \llbracket 1, M \rrbracket \quad (55)$$

be an enumeration of $\mathcal{E}_N = \{(m, n) \mid m, n \geq 0, m + n \leq N\}$.

Consider the polynomials $(h_1, \dots, h_M)$ in $\text{Pol}_N(\mathbb{R}^2)$ defined by

$$\forall (m, n) \in \mathcal{E}_N, \quad h_{\pi(m, n)}(v, x) = v^m H_n(x). \quad (56)$$

The family $(h_1, \dots, h_M)$ is a basis of $\text{Pol}_N(\mathbb{R}^2)$. The following theorem gives an explicit formula for the Hermite moments.

Theorem 3.2 (Hermite moments). *Let $p \in \text{Pol}_N(\mathbb{R}^2)$ and $t \in [0, T]$. Let $\vec{p} \in \mathbb{R}^M$ denote the coordinates of p in the basis $(h_1, \dots, h_M)$. Then, for every $n \in \llbracket 0, N \rrbracket$,*

$$\ell_n = (h_1(V_0, X_0), \dots, h_M(V_0, X_0)) e^{TG} \mathbf{e}_{\pi(0, n)} \quad (57)$$

where G is the matrix of the infinitesimal generator $\mathcal{G}$, viewed as an endomorphism of $\text{Pol}_N(\mathbb{R}^2)$, in the basis $(h_1, \dots, h_M)$.

Proof. The proof is a direct application of a result from [6] (Section 3). ■

Remark 3.8. The matrix G belongs to $\mathbb{R}^{M \times M}$. It was calculated explicitly in [1]. It has at most 7 nonzero entries in each column. Thus G is sparse: if R_N denotes its number of nonzero entries, then the proportion R_N/M^2 tends to 0 as N tends to infinity, since $R_N \leq 7M$.

3.6.5 Numerical Results

Figure 19 shows the first Hermite moments ℓ_n and the first Fourier coefficients f_n .

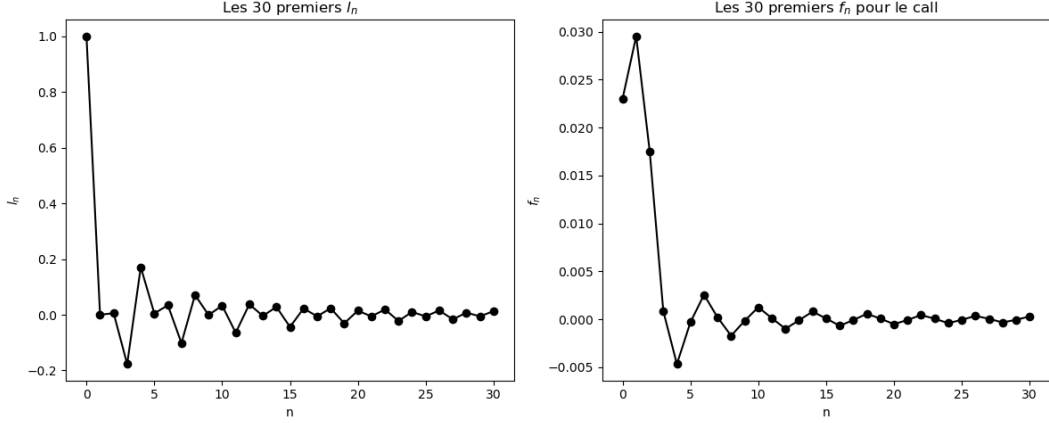


Figure 19: Hermite moments (left) and Fourier coefficients for the call (right). We choose $V_0 = 0.04$, $X_0 = 0$, $v_{\min} = 10^{-4}$, $v_{\max} = 0.08$, $\sigma = 1$, $\rho = -0.5$, $\kappa = 0.5$, $\theta = 0.04$, $\delta = 0$, $k = 0$, $T = 1/12$ and $r = 0$.

Figure 20 shows the call price π_f^N as a function of N . We observe that π_f^N converges to the price obtained by Monte Carlo.

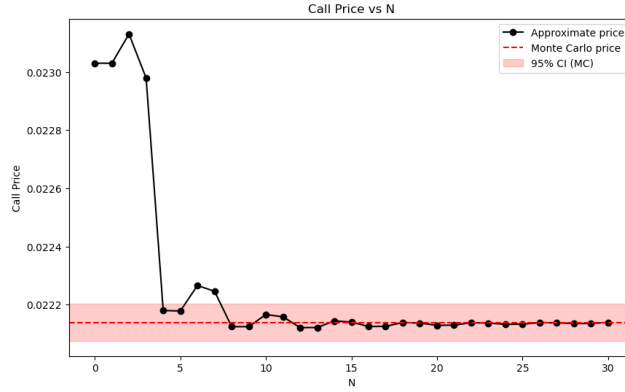


Figure 20: Call price. The chosen parameters are the same as in Figure 19. For Monte Carlo, we chose $m = 10^6$.

Remark 3.9. *These graphs already appear in [1], but we reproduced them to validate our implementation.*

3.6.6 Volterra Version

In this section, we construct the Volterra version of (46). This is not done in [1]. Let

$$K(t) = \sum_{i=1}^d \gamma_i e^{-\rho_i t}.$$

be a Prony series. To simplify the notation, we choose $d = 2$, meaning that we consider two factors V^1 and V^2 . However, everything that follows can be written for an arbitrary number of factors. We now consider the process (V_t, X_t) defined by

$$\begin{cases} V_t = V_0 + \int_0^t K(t-s) \left(\kappa(\theta - V_s) ds + \sigma \sqrt{Q(V_s)} dW_{1s} \right) \\ dX_t = (r - \delta - V_t/2) dt + \rho \sqrt{Q(V_t)} dW_{1t} + \sqrt{V_t - \rho^2 Q(V_t)} dW_{2t}. \end{cases} \quad (58)$$

We can write $V_t = V_0 + \gamma^\top \mathbf{V}_t$, with $\mathbf{V}_t = (V_t^1, V_t^2)$, where the V^i satisfy

$$dV_t^i = -\rho_i V_t^i dt + \kappa(\theta - V_t)dt + \sigma\sqrt{Q(V_t)}dW_{1t} \quad (59)$$

This yields the 3-dimensional equation

$$d\mathbf{Y}_t = b(\mathbf{V}_t)dt + \sigma(\mathbf{V}_t)dW_t \quad (60)$$

where $W_t = (W_{1t}, W_{2t})^\top$ is a 2-dimensional Brownian motion, $\mathbf{Y}_t = (V_t^1, V_t^2, X_t)$, and $b : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ and $\sigma : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ are defined by

$$b(\mathbf{v}) = \begin{pmatrix} -\text{diag}(\boldsymbol{\rho})\mathbf{v} + \kappa(\theta - V_0 - \gamma^\top \mathbf{v})\mathbf{1} \\ r - \delta - (V_0 + \gamma^\top \mathbf{v}) \end{pmatrix} \quad \text{and} \quad \sigma(\mathbf{v}) = \begin{pmatrix} \sigma\sqrt{Q(V_0 + \gamma^\top \mathbf{v})} & 0 \\ \sigma\sqrt{Q(V_0 + \gamma^\top \mathbf{v})} & 0 \\ \rho\sqrt{Q(V_0 + \gamma^\top \mathbf{v})} & \sqrt{V_0 + \gamma^\top \mathbf{v} - \rho^2 Q(V_0 + \gamma^\top \mathbf{v})} \end{pmatrix}.$$

Thus, the infinitesimal generator of (60) is

$$\mathcal{G}f(\mathbf{v}, x) = b(\mathbf{v})^\top \nabla f(\mathbf{v}, x) + \frac{1}{2} \text{Tr}(a(\mathbf{v}) \nabla^2 f(\mathbf{v}, x)) \quad (61)$$

where

$$a(\mathbf{v}) = \sigma(\mathbf{v})\sigma^\top(\mathbf{v}) = \begin{pmatrix} \sigma^2 Q(V_0 + \gamma^\top \mathbf{v}) & \sigma^2 Q(V_0 + \gamma^\top \mathbf{v}) & \rho\sigma Q(V_0 + \gamma^\top \mathbf{v}) \\ \sigma^2 Q(V_0 + \gamma^\top \mathbf{v}) & \sigma^2 Q(V_0 + \gamma^\top \mathbf{v}) & \rho\sigma Q(V_0 + \gamma^\top \mathbf{v}) \\ \rho\sigma Q(V_0 + \gamma^\top \mathbf{v}) & \rho\sigma Q(V_0 + \gamma^\top \mathbf{v}) & V_0 + \gamma^\top \mathbf{v} \end{pmatrix}. \quad (62)$$

The infinitesimal generator $\mathcal{G}$ preserves $\text{Pol}_N(\mathbb{R}^3)$ and can therefore be viewed as an endomorphism of this space when restricted to it. We again choose an enumeration $\pi : \mathcal{E}_N \rightarrow \llbracket 1, M \rrbracket$, now with

$$\mathcal{E}_N = \{(m_1, m_2, n) \mid m_1, m_2, n \geq 0, m_1 + m_2 + n \leq N\}$$

and $M = \binom{N+3}{3}$. Consider the family $(h_1, \dots, h_M)$ given by

$$\forall (m_1, m_2, n) \in \mathcal{E}_N, \quad h_{\pi(m_1, m_2, n)}(v_1, v_2, x) = v_1^{m_1} v_2^{m_2} H_n(x). \quad (63)$$

The following theorem is the analogue of Theorem 3.2 for the Volterra version of the Jacobi model.

Theorem 3.3 (Hermite moments). *Let $p \in \text{Pol}_N(\mathbb{R}^3)$ and $t \in [0, T]$. Let $\vec{p} \in \mathbb{R}^M$ denote the coordinates of p in the basis $(h_1, \dots, h_M)$. Then, for every $n \in \llbracket 0, N \rrbracket$,*

$$\ell_n = (h_1(V_0^1, V_0^2, X_0), \dots, h_M(V_0^1, V_0^2, X_0)) e^{TG} \mathbf{e}_{\pi(0,0,n)} \quad (64)$$

where G is the matrix of the infinitesimal generator $\mathcal{G}$, viewed as an endomorphism of $\text{Pol}_N(\mathbb{R}^3)$, in the basis $(h_1, \dots, h_M)$.

Proof. The proof is again a direct application of the result in [6] (Section 3). ■

We now have all the ingredients needed to calculate the approximate option price π_f^N in this new model, allowing us to perform the pricing numerically. The results are given in the following section.

Figure 21 shows the call price π_f^N as a function of N in this new model.

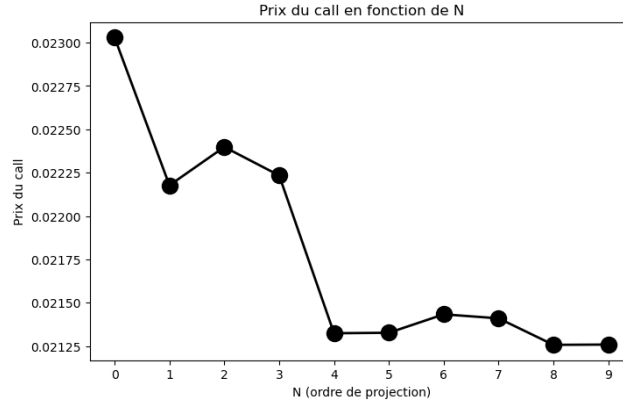


Figure 21: Call price. The chosen parameters are $V_0 = 0.04$, $V_0^1 = V_0^2 = 0$, $X_0 = 0$, $v_{\min} = 10^{-4}$, $v_{\max} = 0.08$, $\sigma = 1$, $\rho = -0.5$, $\theta = 0.04$, $\delta = 0$, $r = 0$, $k = 0$ and $T = 1/12$.

The coefficients in Figure 21 were chosen so as not to depart too far from the model used in Figure 20. It is therefore natural for the prices to be similar.

4 Conclusion

This work allowed me to explore many aspects of stochastic calculus. First, we attempted to extend the results of [7]. In particular, we sought to pass to the limit in Theorem 2.1. Unfortunately, this attempt was unsuccessful. We nevertheless managed to generalise less ambitious results, such as adding a constant to the Volterra process and extending the results to a d -dimensional Volterra process. In the second part, we carried out many simulations. The initial objective was to observe numerically the domain $\mathcal{D}$ from Theorem 2.1. This led us to study several simulation schemes. In this context, I discovered the method of splitting the infinitesimal generator. We then applied it to the Heston model to obtain a financial application, which in turn led us to consider another model, the Jacobi model.

My internship lasted only two months, which is rather short for carrying out truly comprehensive work. This work could be continued in several ways. For example, some simulations could be refined or extended to other models. Likewise, the theoretical aspects could be investigated in greater depth.

In conclusion, this research internship enabled me to learn a great deal, both mathematically and about research methodology. Mathematical research is a field that interests me, and I am considering, to some extent—that is, with about a 30% chance—pursuing a PhD after completing my second-year master’s degree. This internship showed me that a PhD is a genuine possibility because I enjoy this field. Finally, I would like to thank Aurélien Alfonsi for his supervision, teaching and kindness throughout the internship. I am very grateful to have had the opportunity to work under his supervision.

A Additional Results

A.1 Proof of Theorem 2.1

Let $b : \mathbb{R}^N \rightarrow \mathbb{R}^N$ and $\sigma : \mathbb{R}^N \rightarrow \mathbb{R}^{N \times d}$ be continuous functions with at most linear growth. We consider the SDE

$$d\mathbf{X}_t = b(\mathbf{X}_t)dt + \sigma(\mathbf{X}_t)dW_t, \quad \mathbf{X}(0) = \mathbf{x}. \quad (65)$$

Definition A.1 (Invariance). *A closed convex set K is said to be invariant if, for every $x \in K$, there exists a weak solution $\mathbf{X}$ of (65) that remains in K over time.*

Let $d(\mathbf{x}) = \inf_{\mathbf{y} \in K} |\mathbf{x} - \mathbf{y}|$. The function d is continuously differentiable almost everywhere, and

$$d'(\mathbf{x}) = \frac{\mathbf{x} - \pi_K(\mathbf{x})}{|\mathbf{x} - \pi_K(\mathbf{x})|}. \quad (66)$$

Lemma A.1 (Viability lemma). *Let K be a closed convex set, and set $K^c = \mathbb{R}^N \setminus K$. Then K is invariant if and only if*

$$\sigma(\pi_K(\mathbf{x}))^\top d'(\mathbf{x}) = 0 \quad \text{and} \quad L_K d(\mathbf{x}) \leq 0 \quad \text{for almost every } x \in K^c \quad (67)$$

where $L_K f(\mathbf{x}) = \frac{1}{2} \text{Tr}(\sigma(\pi_K(\mathbf{x}))\sigma(\pi_K(\mathbf{x}))^\top f''(\mathbf{x})) + \langle f'(\mathbf{x}), b(\pi_K(\mathbf{x})) \rangle$.

Proof. The proof is lengthy and constitutes the main result of [5], to which we refer the reader. ■

The following corollary applies Lemma A.1 to $K = \mathbb{R}_+^N$.

Corollary A.1. *The invariance conditions (67) are equivalent to*

$$\forall \mathbf{z} \in \mathbb{R}_+^N, \quad z_i = 0 \implies \sigma(\mathbf{z})^\top \mathbf{e}_i = 0 \quad \text{and} \quad b_i(\mathbf{z}) \geq 0$$

and

$$\sigma(0) = 0, \quad b_i(0) \geq 0 \quad \forall i \in \llbracket 1, N \rrbracket.$$

Proof. This application of Lemma A.1 is mentioned in [5], but is not proved there. We therefore provide a proof. Suppose that $K = \mathbb{R}_+^N$, and assume that $\sigma(\pi_K(\mathbf{x}))^\top d'(\mathbf{x}) = 0$ and $L_K d(\mathbf{x}) \leq 0$ for every $\mathbf{x} \in K^c$. Let $\mathbf{z} \in \partial K \setminus \{0\}$ be such that $z_i = 0$. Set $\mathbf{z}(t) = \mathbf{z} - t\mathbf{e}_i$ for $t > 0$. Then, for every $t > 0$, $\mathbf{z}(t) \in K^c$, and hence $\sigma(\pi_K(\mathbf{z}(t)))^\top d'(\mathbf{z}(t)) = 0$ and

$$L_K d(\mathbf{z}(t)) = \frac{1}{2} \text{Tr}(\sigma(\pi_K(\mathbf{z}(t)))\sigma(\pi_K(\mathbf{z}(t)))^\top d''(\mathbf{z}(t))) + \langle d'(\mathbf{z}(t)), b(\pi_K(\mathbf{z}(t))) \rangle \leq 0.$$

Now, $\pi_K(\mathbf{z}(t)) = \mathbf{z}$ and $d(\mathbf{z}(t)) = t$ for every $t > 0$. Differentiating with respect to t gives

$$z'(t)^\top d'(\mathbf{z}(t)) = -\mathbf{e}_i^\top d'(\mathbf{z}(t)) = 1, \quad d''(\mathbf{z}(t)) = 0$$

Then $L_K d(\mathbf{z}(t)) = \langle -\mathbf{e}_i, b(\mathbf{z}) \rangle = -b_i(\mathbf{z}) \leq 0$, so $b_i(\mathbf{z}) \geq 0$, and $\sigma(\mathbf{z})^\top (-\mathbf{e}_i) = 0$.

By continuity of b and σ , taking a sequence $(z_n)_{n \in \mathbb{N}}$ in $\partial K \setminus \{0\}$ such that $[z_n]_i = 0$ for every $n \in \mathbb{N}$ and z_n tends to 0, passage to the limit gives $\sigma(0) = 0$ and $b_i(0) \geq 0$.

Conversely, suppose that, for every $\mathbf{z} \in \partial K \setminus \{0\}$ such that $z_i = 0$, we have $\sigma(\mathbf{z})^\top \mathbf{e}_i = 0$ and $b_i(\mathbf{z}) \geq 0$, as well as $\sigma(0) = 0$ and $b_i(0) \geq 0$. Let $\mathbf{x} \in K^c$.

If $\pi_K(\mathbf{x}) \neq 0$, i.e. $\mathbf{x}$ has at least one strictly positive entry, say $z_i > 0$, then there exists $t_0 > 0$ such that $\mathbf{x} = \pi_K(\mathbf{x}) - t_0 \mathbf{e}_i$. The preceding calculations give $-\mathbf{e}_i d'(\mathbf{x}) = 1$ and $d''(\mathbf{x}) = 0$. Combining this with the assumption applied to $\pi_K(\mathbf{x})$ immediately yields $\sigma(\pi_K(\mathbf{x}))^\top d'(\mathbf{x}) = 0$ and $L_K d(\mathbf{x}) \leq 0$.

If $\pi_K(\mathbf{x}) = 0$, the identities $\sigma(0) = 0$ and $b_i(0) \geq 0$ directly give $\sigma(\pi_K(\mathbf{x}))^\top d'(\mathbf{x}) = 0$ and $L_K d(\mathbf{x}) \leq 0$. $\blacksquare$

A.2 M-matrices

Definition A.2 (Spectral radius). *Let $A \in \mathbb{R}^{N \times N}$. The spectral radius of A , denoted by $\rho(A)$, is*

$$\rho(A) := \max \{ |\lambda| \mid \lambda \in \text{Sp}(A) \}$$

By Gelfand's theorem (see [9]), the spectral radius of a matrix A is also given by

$$\rho(A) = \inf_{n \in \mathbb{N}^*} \|A^n\|^{1/n} = \lim_{n \in \mathbb{N}^*} \|A^n\|^{1/n}. \quad (68)$$

We introduce another result giving a convergence criterion for series of the form $\sum_{n \geq 0} z^n A^n$.

Proposition A.1. *Let $A \in \mathbb{R}^{N \times N}$ and $z \in \mathbb{C}$. Then $\sum_{n \geq 0} z^n A^n$ converges if and only if $|z| < \frac{1}{\rho(A)}$.*

Proof. Suppose that $|z| < \frac{1}{\rho(A)}$, and let $\varepsilon > 0$. Since $\rho(A) = \lim_{n \rightarrow \infty} \|A^n\|^{1/n}$ by Gelfand's theorem, there exists $n_0 \geq 1$ such that, for every $n \geq n_0$, $\|A^n\|^{1/n} \leq \rho(A) + \varepsilon$, and hence $\|A^n\| \leq (\rho(A) + \varepsilon)^n$. Therefore,

$$|z|^n \|A^n\| \leq [|z|(\rho(A) + \varepsilon)]^n \quad \forall n \geq n_0.$$

Since $|z|\rho(A) < 1$, ε can be chosen small enough that $|z|(\rho(A) + \varepsilon) < 1$. Thus $\sum_{n \geq 0} z^n A^n$ converges normally and therefore converges.

Conversely, suppose that $\sum_{n \geq 0} z^n A^n$ converges. Its general term therefore tends to 0. Since $\rho(A) = \inf_{n \in \mathbb{N}^*} \|A^n\|^{1/n}$, for every $n \in \mathbb{N}^*$,

$$|z|^n \|A\|^n \geq |z|^n \rho(A)^n.$$

The left-hand side tends to 0, and hence so does the right-hand side. By the convergence criterion for a geometric sequence, $|z|\rho(A) < 1$, i.e. $|z| < \frac{1}{\rho(A)}$. $\blacksquare$

Definition A.3 (M-matrix). *Let $\mathcal{R}(z)$ denote the real part of a complex number $z \in \mathbb{C}$. Let $A = (a_{ij}) \in \mathbb{R}^{N \times N}$. The matrix A is an M-matrix if*

1. $\forall i \neq j, a_{ij} \leq 0$
2. $\forall \lambda \in \text{Sp}(A), \mathcal{R}(\lambda) > 0$.

The following proposition is taken from [4].

Proposition A.2 (Nonnegativity of the entries of the inverse of an M-matrix). *Let A be an M-matrix. Then A is invertible and, for every $i, j \in \llbracket 1, N \rrbracket$, $[A^{-1}]_{ij} \geq 0$.*

Proof. We proceed in two steps.

Step 1. We show that an M-matrix A can be written as $A = sI_N - B$, with $b_{ij} \geq 0$ for every $i, j \in \llbracket 1, N \rrbracket$ and $s > \rho(A)$. Indeed, let s be a real number such that $s > \max_{i \in \llbracket 1, N \rrbracket} a_{ii}$ and $s > \rho(A)$. With $B = sI_N - A$, we have $b_{ij} = s\delta_{ij} - a_{ij}$. Thus $b_{ii} = s - a_{ii} \geq 0$ by the

choice of s , while, for $i \neq j \in \llbracket 1, N \rrbracket$, $b_{ij} = -a_{ij} \geq 0$ because the off-diagonal entries of the M-matrix A are nonpositive.

Step 2. The matrix A is invertible because all its eigenvalues are nonzero. Moreover,

$$A = sI_N - B = s[I_N - B/s].$$

Since $s > \rho(B)$, we have $\rho(B/s) < 1$. Therefore, applying Proposition A.1 with $z = 1$ and B/s , the series $\sum \left(\frac{B}{s}\right)^n$ converges, and

$$A^{-1} = \frac{1}{s} \sum_{n \geq 0} \left(\frac{B}{s}\right)^n.$$

Since the entries of B are nonnegative, it follows readily that the entries of A^{-1} are also nonnegative. ■

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